



Integrable Deformations of Cluster Maps of Type D_{2N}

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Abstract

In this paper, we extend one of the main results from our joint work [12] with Hone and Mase, in which we studied a deformed type D_4 map, to the general case of type D_{2N} for $N \geq 3$. This can be achieved through a “local expansion” operation, introduced in our joint work [7] with Grabowski and Hone. This operation involves inserting a specific subquiver into the quiver arising from the Laurentification of the deformed type D_4 map. This insertion yields a new quiver, obtained through the Laurentification of the deformed type D_6 map and thus enables systematic generalization to higher ranks D_{2N} . We also study the degree growth of the deformed type D_{2N} map via the tropical method and conjecture that, for each N , the deformed map is integrable, as indicated by the algebraic entropy test, a criterion for detecting integrability in discrete dynamical systems.

Keywords Cluster algebra · Quiver · Integrable map · Laurent property · Algebraic entropy

Mathematics Subject Classification 39A36 · 13F60

1 Introduction

Cluster algebras are commutative algebras which were introduced by Fomin and Zelevinsky [2]. These algebras are defined by a collection of distinguished generators, *cluster variables* which are produced through the iterative process of *mutation*. Performing mutation μ_j on a *cluster* $\mathbf{x} = (x_1, \dots, x_n)$, n -tuple of cluster variables, replaces the variable x_j with a new variables x'_j given by the following difference equation,

$$x'_j x_j = F(x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)$$

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where $x'_j = \mu_j(x_j)$ and F is a subtraction-free polynomial in initial cluster variables. In the cluster algebra, this equation is known as *exchange relation*, and it is determined by the *quiver*, Q , that is a finite directed graph, consisting of no loops or 2-cycles. Moreover the quiver itself transforms under mutation, together with cluster variables. Thus more precisely, a cluster algebra is a collection of *seeds*, where each seed is formed by a set of distinguished cluster variables and associated quiver. These seeds are induced from the initial seed $(\mathbf{x}, Q)$ by recursively applying the mutations.

The exchange relation is said to possess the *Laurent property*, that is, every variable induced by the exchange relation can be expressed as a Laurent polynomial in initial variables. In other words, new cluster variables induced by any composition of mutations $\mu_{i_n} \cdot \mu_{i_{n-1}} \cdots \mu_{i_2} \cdot \mu_{i_1}$ are elements of the Laurent polynomial ring $\mathbb{Z}_{\geq 0}[x_1^{\pm}, x_2^{\pm}, \dots, x_n^{\pm}]$. This is a key distinguished feature of cluster algebras.

In a particular case, if there exists a composition of mutations that fixes the quiver, then it can be regarded as a birational map which maps between clusters, $\varphi : \mathbf{x} \rightarrow \mathbf{x}' = (x'_1, x'_2, \dots, x'_n)$, which is referred to as *cluster map*. This enables the description of discrete dynamical systems within the framework of cluster algebra, which leads to the study of discrete integrable systems [6, 15].

For instance, a non-linear recurrence relation of the form

$$x_{n+2}x_n = 1 + x_{n+1} \quad (1)$$

is known as the *Lyess recurrence relation*. The recurrence is equivalent to the iteration of the birational map given by

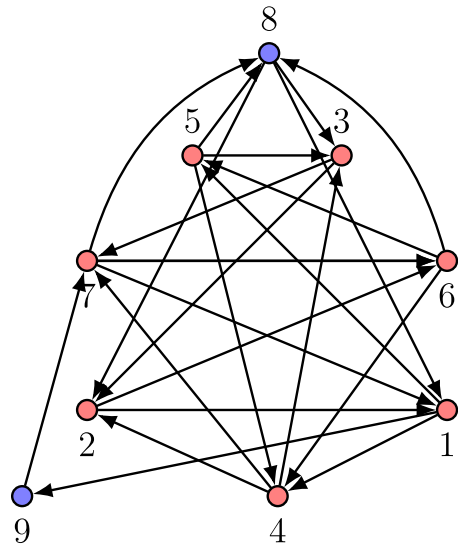
$$\varphi : (x_1, x_2) \rightarrow \left(x_2, \frac{1 + x_2}{x_1} \right) \quad (2)$$

This map can be constructed by a composition of mutations in the cluster algebra with quiver which is identical to the Dynkin type A_2 . This map preserves the symplectic 2-form

$$\omega = \frac{1}{x_1 x_2} dx_1 \wedge dx_2 \quad (3)$$

and there exists a symmetric function (first integral) which is invariant under the action of the map. Thus, it is a Liouville integrable map whose iteration represents a discrete integrable system. One can observe that the orbit of recurrence relation is periodic with period 5, i.e. $(x_{n+5} = x_n)$. This is the simplest example of Zamolodchikov periodicity, that is, the periodicity of the orbit of birational maps for which iterates returns to initial data after a fixed number of evolutions. This periodicity was first observed by Zamolodchikov [21] in the dynamics induced by the difference equation known as the *Y-system*, which provides solutions to the Bethe ansatz equations associated with conformal field theories of the ADE scattering type. This Y-system was later realized within the framework of cluster algebra by introducing *coefficient variable* defined over a semifield. Subsequently, Fomin and Zelevinsky showed that both dynamics of coefficient variable and cluster variable exhibit same periodicity in the cluster algebras whose quiver has same structure as the finite Dynkin diagram [4].

Fig. 1 Quiver associated with special case of Somos-7 recurrence (6)



The following parameter modification on the recurrence relation

$$x_{n+2}x_n = b + ax_{n+1} \quad (4)$$

does preserve the symplectic structure and integrability of the map. In other words, one can generalise the integrable cluster map by imposing arbitrary parameters in the exchange relation. This deformation provides the motivation behind the development of the results presented by Hone and Kouloukas in [11]. However, in exchange for admitting deformation of the cluster map, the iteration no longer satisfies the Laurent property. This implies that the map cannot be defined in cluster algebras. One way to restore this property is to lift the deformed map to a higher dimensional space where the Laurent property holds. This method is so called Laurentification (a term coined by Hamad et al. in [9]), which is established by constructing a variable transformation through the p -adic method introduced by Kanki in [13] which is analogous to the singularity confinement test in discrete systems [8]. Through the following variable transformation,

$$x_n = \frac{\tau_{n+5}\tau_n}{\tau_{n+2}\tau_{n+3}} \quad (5)$$

the recurrence relation can be rewritten as

$$\tau_{n+7}\tau_n = a\tau_{n+1}\tau_{n+6} + b\tau_{n+3}\tau_{n+4} \quad (6)$$

which can be realised via mutations in cluster algebra associated with the quiver shown in Fig. 1. This means that the parametric map can be generated from the cluster variables in cluster algebra of higher rank.

There is another form of the cluster map which generates the same sequence of cluster variables as the Lyness recurrence. The iteration of the map is equivalent to

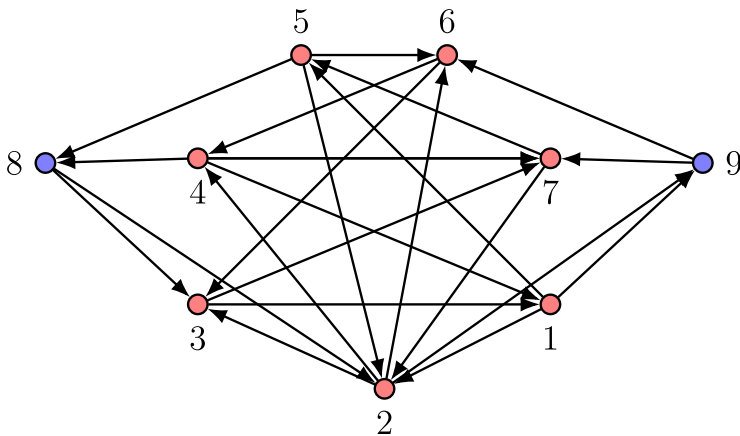


Fig. 2 The extended quiver Q_{A_2} constructed through Laurentification of (8)

the system of difference equations

$$\begin{aligned} x_{1,n+1}x_{1,n} &= 1 + x_{2,n} \\ x_{2,n+1}x_{2,n} &= 1 + x_{1,n+1} \end{aligned} \quad (7)$$

Choosing a different deformation (compared to (4))

$$\begin{aligned} x_{1,n+1}x_{1,n} &= 1 + a_1x_{2,n} \\ x_{2,n+1}x_{2,n} &= 1 + a_2x_{1,n+1} \end{aligned} \quad (8)$$

gives rise to a deformed map which preserves the same symplectic form as the Lyness map and yields a discrete integrable system. Under Laurentification, using a variable transformation consisting a different form from that of the Lyness map, the deformed map is lifted to a cluster algebra whose quiver is identical to the quiver in Fig. 1 but with its frozen vertices connected differently (shown in Fig. 2).

In the one of previous work [7], we showed that the quiver arising from deformation of type A_4 cluster, can be arranged in the form of Q_{A_4} ((b) in Fig. 3). This configuration has structure that allows the Q_{A_2} to be extended into Q_{A_4} as shown in Fig. 3 by insertion of particular quiver, referred to as local expansion, illustrated in the Fig. 4.

By applying this expansion recursively, one can build class of quivers $Q_{A_{2N}}$ (represented in Fig. 5), together with an associated family of parametric cluster maps $\psi_{A_{2N}}$. As a result, the Laurent property of the deformed type A_{2N} maps is restored by defining them on corresponding cluster algebras of higher rank.

In this paper, we follow a similar approach used in [7] to construct a family of quivers that yields a two-parameter cluster map associated with the deformation of a type D_{2N} cluster map. The paper is outlined as follows.

In Sect. 2, we begin by reviewing a brief background on cluster algebras and consider the cluster algebra of type D_4 to introduce features such as the Laurent property,

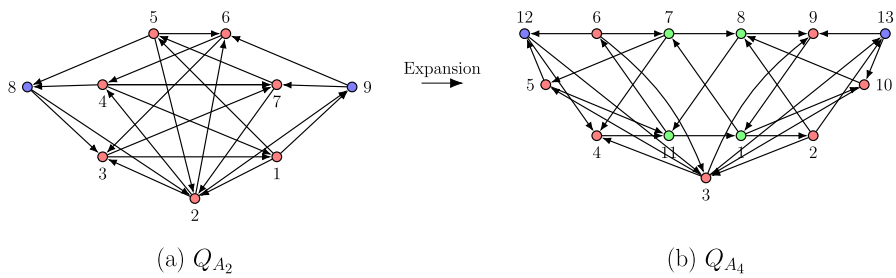


Fig. 3 Extension from Q_{A_2} to Q_{A_4} . Green nodes are new vertices which are inserted in Q_{A_2} in the form of local expansion (shown in the Fig. 4)

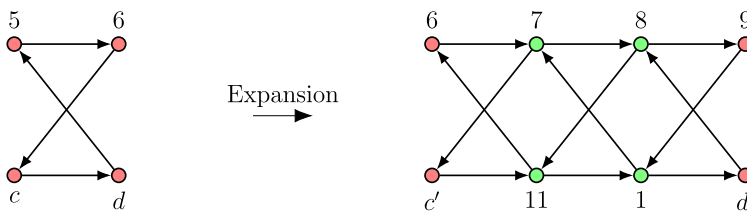


Fig. 4 Local expansion of the subquiver in Q_{A_2} . The nodes $c = \{3, 4\}$, $d = \{1, 7\}$, $c' = \{4, 5\}$ and $d' = \{2, 10\}$

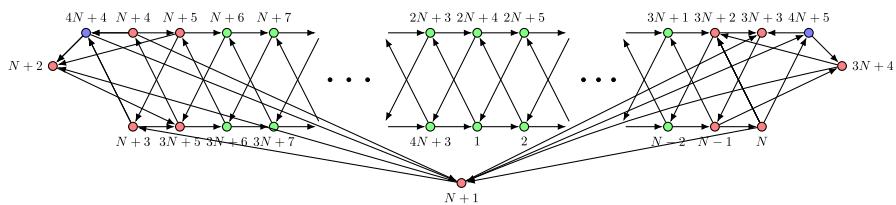


Fig. 5 Quiver, $Q_{A_{2N}}$ associated to deformation of type A_{2N} cluster map [7]. The red vertices indicate the nodes that already existed in Q_{A_4} while the green vertices represent those newly included through local expansion

mutation periodicity and Zamolodchikov periodicity. We then review the discrete integrable systems in the framework of cluster algebras.

In Sect. 3, we review the integrable deformation of the type D_4 cluster map from [12]. Under the parameter conditions for integrability, the map reduces to two distinct parametric Liouville integrable maps on the plane, each preserving a different first integral. Both maps admit Laurentifications, which are inequivalent to each other, that lift to cluster algebras of rank 8. The two quivers share the same configuration of mutable nodes, but the frozen nodes are connected differently.

In Sect. 4 we examine the deformation of the cluster map associated with type D_6 and show that it admits Laurentification, which lifts to a cluster algebra of rank 12 with a quiver consisting of two frozen nodes. Using the max-plus form of the exchange relations, we determine the degree growth and compute the *algebraic entropy*, a standard criterion for integrability.

Subsequently, in Sect. 5, we compare the quiver induced in the type D_6 case with one of the extended quivers arising from the deformation in cluster algebra of type D_4 and show that they are related through the insertion of specific subquiver in the form of a local expansion. Iterating this expansion yields a quiver with $4N$ nodes with two frozen nodes, which corresponds to the Laurentification of the deformed type D_{2N} map.

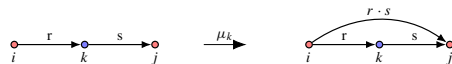
In Sect. 6, we prove that the extended quiver constructed in the previous section is mutation periodic up to a specific permutation of the labelling of nodes, thereby defining the cluster map associated with the deformed type D_{2N} map. We further show that, for all $N > 1$, the degree growth of the map is quadratic.

2 Preliminaries

Cluster algebra of rank n is generated by distinguished generators, known as *cluster variables*, induced by applying mutation recursively starting from the initial seed, $(\mathbf{x}, Q)$ consisting of a set of cluster variables $\mathbf{x} = (x_1, \dots, x_n)$ and a quiver Q , that is, a finite directed graph with n vertices, which does not contain any loops (1-cycles) or 2-cycles. The mutation μ_k applies to each object in the seed separately, $\mu_k(\mathbf{x}, Q) = (\mu_k(\mathbf{x}), \mu_k(Q)) = (\mathbf{x}', Q')$, following individual transformation rules. Let us first consider *quiver mutation*.

Quiver mutation: The mutation $\mu_k(Q)$ at the node k gives rise to a new quiver Q' by taking following rule as shown below,

1. For each path $i \xrightarrow{r} k \xrightarrow{s} j$ in Q , multiply the number of incoming and outgoing arrows at the node k and insert the resulting number of edges between i and j ,



2. Reverse all edges which are incident to node k
3. Remove any 2-cycles formed by the step 1.

The quiver with n nodes can be encoded by $n \times n$ skew-symmetric matrix, $B \in \text{Mat}(\mathbb{Z})$ in which each entry b_{ij} represents the number of arrows from i to j i.e.

$$b_{ij} = \begin{array}{c} i \\ \bullet \end{array} \longrightarrow \begin{array}{c} j \\ \bullet \end{array}$$

The corresponding matrix is known as *exchange matrix*. The quiver mutation μ_k at the node k , can be formulated by using the entries of the exchange matrix B as shown below.

Matrix mutation $\mu_k(B)$: The new exchange matrix $B' = \mu_k(B)$ is given by

$$(\mu_k(B))_{ij} = (B')_{ij} = b'_{ij} = \begin{cases} -b_{ij} & \text{if } i = k \text{ or } j = k \\ b_{ij} + \frac{1}{2} (|b_{ik}| b_{kj} + b_{ik} |b_{kj}|) & \text{otherwise} \end{cases} \quad (9)$$

Together with quiver (or matrix mutation), cluster variable x_k in the cluster $\mathbf{x}$ is updated to a new variable, which is given by the equation below.

Cluster mutation $\mu_k(\mathbf{x})$: The *cluster mutation* of $\mathbf{x}$ in direction k , $\mu_k(\mathbf{x}) = (x_1, \dots, x_{k-1}, x'_k, \dots, x_n)$, give rise to a new cluster variable x'_k which is given by the following expression,

$$\mu_k(x_k) = x'_k = \frac{1}{x_k} \left(\prod_{\substack{j=1 \\ b_{jk} > 0}}^n x^{b_{jk}} + \prod_{\substack{j=1 \\ b_{jk} < 0}}^n x^{-b_{jk}} \right) \quad (10)$$

This expression is known as the (coefficient free) *exchange relation*. Note that the mutation is an involution in the sense that the quiver remains unchanged when the same mutation is applied twice in the same direction i.e. $\mu_k \cdot \mu_k(\mathbf{x}, Q) = \mu_k^2(\mathbf{x}, Q) = (\mathbf{x}, Q)$. The sequence of mutations starting from the initial seed $(\mathbf{x}_0, B_0)$,

$$(\mathbf{x}_0, B_0) \xrightarrow{\mu_{i_1}} (\mathbf{x}_1, B_1) \xrightarrow{\mu_{i_2}} (\mathbf{x}_2, B_2) \xrightarrow{\mu_{i_3}} \dots \xrightarrow{\mu_{i_m}} (\mathbf{x}_m, B_m)$$

produces new cluster variables and quivers, resulting in construction of the cluster algebra.

Definition 2.1 (Cluster algebra) Let $\mathcal{F}$ be the field of rational functions in n independent variables $x_1, \dots, x_n$ over $\mathbb{C}$ and set $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{F}^n$. Let B be an $n \times n$ integer skew-symmetric matrix. Then cluster algebra of rank n , $\mathcal{A}(\mathbf{x}, B)$, is the subring of $\mathcal{F}$ generated by the set of distinguished cluster variables in all seeds that are reachable by sequences of mutations starting from the initial seed.

One can extend the notion of cluster algebra by allowing the exchange matrix to be *skew-symmetrizable* matrix, and by incorporating non-mutable variables, called *frozen variables*, in cluster, together with their corresponding *frozen nodes* in the quiver.

The definition of a skew-symmetrizable matrix is given below.

Definition 2.2 An $n \times n$ integer matrix B is skew-symmetrizable if there exists an integer diagonal matrix D , called a skew-symmetrizer, such that BD is skew-symmetric i.e. $(BD)^T = -BD$.

This generalization allows one to consider cluster algebras whose exchange matrices are skew-symmetrizable but not necessarily skew-symmetric. For example, a cluster algebra of Dynkin type B_2 has exchange matrix

$$\begin{pmatrix} 0 & 2 \\ -1 & 0 \end{pmatrix}$$

which is skew-symmetrizable with diagonal matrix $D = \text{diag}(2, 1)$.

As for the latter extension, let us consider a cluster algebra of rank n with initial cluster $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and initial quiver Q with n mutable nodes. To generalise the structure, we introduce the m frozen variables $\tilde{x}_{n+1}, \tilde{x}_{n+2}, \dots, \tilde{x}_{n+m}$ and corresponding m frozen nodes. As the name suggests, these variables and nodes

never mutate. The frozen variables are added to cluster, forming an *extended cluster* $\tilde{\mathbf{x}} = (x_1, x_2, \dots, x_n, \tilde{x}_{n+1}, \dots, \tilde{x}_{n+m})$ and the quiver is extended to $\tilde{Q}$ with $n + m$ nodes, where m nodes are designated as frozen. Then the extended cluster algebra can be built from the extended initial seed $(\tilde{\mathbf{x}}, \tilde{Q})$ by applying a sequence of mutations

$$x'_k x_k = \alpha_k \prod_{\substack{j=1 \\ b_{jk} > 0}}^n x_j^{b_{jk}} + \beta_k \prod_{\substack{j=1 \\ b_{jk} < 0}}^n x_j^{-b_{jk}} \quad (11)$$

where the coefficients are

$$\alpha_k = \prod_{\substack{j=n+1 \\ b_{j,k} > 0}}^{n+m} x_j^{b_{j,k}}, \quad \beta_k = \prod_{\substack{j=n+1 \\ b_{j,k} < 0}}^{n+m} x_j^{-b_{j,k}}. \quad (12)$$

Note that mutation rule for the extended quiver is same as previously given. We refer to these more general cluster algebras as *cluster algebras of geometric type*.

Let us consider an example of coefficient free cluster algebra.

Example 2.1 Begin with the initial seed $(\mathbf{x}, B)$, where $\mathbf{x} = (x_1, x_2)$ and the exchange matrix

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

The mutation in the direction 1, $\mu_1(\mathbf{x}, B) = (\mathbf{x}', B')$, yields

$$(x'_1, x'_2) = \left(\frac{1+x_2}{x_1}, x_2 \right), \quad B' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -B.$$

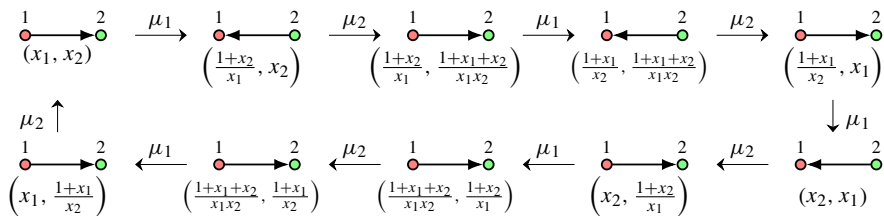
Since the mutation returns the initial seed if we apply the mutation in the same direction as previously, the next mutation should be in direction 2. The mutation $\mu_2(\mathbf{x}') = (x''_1, x''_2)$, gives

$$\begin{aligned} x''_1 &= x'_1 \\ x''_2 &= \frac{1+x'_1}{x'_2} = \frac{1 + \left(\frac{1+x_2}{x_1} \right)}{x_2} = \frac{1+x_1+x_2}{x_1 x_2} \end{aligned}$$

and

$$\mu_2(B') = B$$

Applying mutation in sequence yields the following cluster variables.



One can see that the seed returns to the initial seed after a sequence of 10 mutations. This means that no new cluster variables, except the variables above, can be generated by mutations in any directions. Thus, from the sequence of mutations above, we see that the distinct cluster variables are given by

$$\text{Distinct cluster variables} = \left\{ x_1, x_2, \frac{1+x_2}{x_1}, \frac{1+x_1+x_2}{x_1x_2}, \frac{1+x_1}{x_2} \right\}$$

Hence the associated cluster algebra is given by

$$\mathcal{A}(\mathbf{x}) = \mathbb{C} \left[x_1, x_2, \frac{1+x_2}{x_1}, \frac{1+x_1}{x_2}, \frac{1+x_1+x_2}{x_1x_2} \right]$$

In the example above, we observe that three distinct phenomena arise. Each of these corresponds to features of the cluster algebra and plays an important role in the main result of this paper. The first phenomenon is that every generated cluster variable is expressed as a Laurent polynomial in the initial variables x_i for $i = 1, \dots, n$. This is known as the *Laurent phenomenon*, stated as follows.

Theorem 2.1 (Laurent phenomenon (see [3])) *Every cluster variable generated by cluster mutations is a Laurent polynomial in the initial cluster variables.*

The second phenomenon is that the quiver returns to its initial state in two ways: one is by applying mutation μ_1 followed by the μ_2 (i.e. $\mu_2 \cdot \mu_1$); the other is by applying μ_1 and then perform permutation of the two nodes (i.e. $(1, 2) \cdot \mu_1$). Such a phenomenon is referred to as *mutation periodic* up to permutation, and is stated as follows.

Definition 2.3 (*Mutation periodic*) Let Q be a quiver with n vertices. Then the quiver is *mutation periodic* up to a permutation ρ if there exists a sequence of quiver mutations which is equivalent to applying permutation ρ to the labelling of the quiver Q i.e.

$$\mu_{i_m} \mu_{i_{m-1}} \cdots \mu_{i_2} \mu_{i_1}(Q) = \rho(Q),$$

where $\rho : (1, 2, \dots, N) \rightarrow (\rho(1), \rho(2), \dots, \rho(N))$ is the corresponding permutation.

This notion leads to definition of a certain birational map between cluster seeds,

Definition 2.4 (*Cluster map*) Let $(\mathbf{x}, Q)$ be an initial seed with an initial cluster $\mathbf{x} = (x_1, \dots, x_n)$ and a mutation periodic quiver Q i.e. $\mu_{i_m} \cdots \mu_{i_1}(Q) = \rho(Q)$ for some $m > 0$ and some permutation ρ . Then

$$\varphi = \rho^{-1} \mu_{i_m} \mu_{i_{m-1}} \cdots \mu_{i_2} \mu_{i_1}$$

is a birational map such that

$$\begin{aligned} \varphi : \mathbb{C}^n &\rightarrow \mathbb{C}^n \\ \mathbf{x} = (x_1, \dots, x_n) &\mapsto \mathbf{x}' = (x'_1, \dots, x'_n) \end{aligned} \quad (13)$$

where $\mathbf{x}' = (x'_1, \dots, x'_n)$ is the new cluster given by $\varphi(\mathbf{x})$. Such a map is called a *cluster map*.

This implies that if there exists a composition of mutations that restores the associated quiver (or exchange matrix) after a permutation of nodes, then it can be regarded as a birational map which maps between cluster variables, and is referred to as *cluster map*. This enables the description of discrete dynamical systems within the framework of cluster algebra, which leads to the study of discrete integrable systems.

Finally, the third (and last) phenomenon, which can be observed in Example 2.1, is that the seed returns to the initial seed under the repeated application of the composition of mutations $\mu_2 \mu_1$. In particular, the cluster algebra associated with a finite Dynkin diagram exhibits a remarkable periodicity phenomenon, which is referred to as *Zamolodchikov periodicity*.

Theorem 2.2 (Zamolodchikov periodicity [1, 4, 5]) *Let $\mathcal{A}(\mathbf{x}, Q)$ be a cluster algebra together with a quiver Q corresponding to a finite Dynkin diagram. Let h be the Coxeter number of the corresponding finite Dynkin diagram, and let φ be a composition of cluster mutations such that $\varphi(Q) = Q$. Then there are two cases of periodicity, such as*

- period $\frac{h+2}{2}$

$$\mu^{\frac{h+2}{2}}(\mathbf{x}, Q) = (\mathbf{x}, Q) \quad (14)$$

if associated roots α_i satisfies $w_0(\alpha_i) = -\alpha_i$ where w_0 is the longest element of the corresponding Weyl group,

- period $h + 2$,

$$\mu^{h+2}(\mathbf{x}, Q) = (\mathbf{x}, Q) \quad (15)$$

otherwise.

By combining all three phenomena observed above, we can define a cluster map of the Dynkin diagram that preserves the associated quiver and exhibits Zamolodchikov periodicity.

$$\varphi = (1, 2) \cdot \mu_1 : (x_1, x_2) \rightarrow \left(x_2, \frac{1 + x_2}{x_1} \right), \text{ or}$$

$$\varphi = \mu_2 \mu_1 : (x_1, x_2) \rightarrow \left(\frac{1+x_2}{x_1}, \frac{1+x_1+x_2}{x_1 x_2} \right) \quad (16)$$

satisfying $\varphi^5 = \text{id}$. We remark that although the two maps may appear distinct, they produce the same set of cluster variables.

The iteration of cluster maps gives rise to a discrete dynamical system, which allows one to adopt the notion of Liouville integrable map, introduced in [15], by using the *log-canonical Poisson bracket and (pre) symplectic form*. Both of these structures are compatible with cluster algebras in the sense that they retain their structure under the cluster mutations, i.e.

$$\begin{aligned} (\mathbf{x}, B) &\mapsto \mu_k(\mathbf{x}, B) = (\mathbf{x}', B') = (x'_i) \\ \{x_i, x_j\} = P_{ij} x_i x_j &\rightarrow \{x'_i, x'_j\} = P'_{ij} x'_i x'_j \\ \omega = \sum_{i < j} \frac{b_{ij}}{x_i x_j} dx_i \wedge dx_j &\rightarrow \omega' = \sum_{i < j} \frac{b'_{ij}}{x'_i x'_j} dx_i \wedge dx_j \end{aligned} \quad (17)$$

where $P = (P_{ij})$ is a skew-symmetric matrix, which is directly proportional to inverse of an exchange matrix $B = (b_{ij})$. The two-form ω is referred to as a *presymplectic form* when B is degenerate, and as a *symplectic form* otherwise. Since the cluster map leaves exchange matrix B invariant, the notion of cluster maps can be extended to Poisson (or symplectic) maps. Hence we have the following

Definition 2.5 Let $\mathcal{A}(\mathbf{x}, B)$ be a cluster algebra of rank N and let P be a skew-symmetric matrix of rank $2r < N$ whose entries are the coefficients of the *log-canonical Poisson bracket*, $\{x_i, x_j\} = P_{ij} x_i x_j$. Then the φ is *integrable* if there exists

1. $N - 2r$ Casimir functions C_k , i.e. $\varphi^*(C_k) = C_k$ satisfying $\{C_k, f(\mathbf{x})\} = 0$ for all function $f(\mathbf{x})$
2. r first integrals h_j , $j = 1, \dots, r$, $(\varphi^*(h_j) = h_j)$ such that $\{h_i, h_j\} = 0$

For the non-degenerate case with $\text{rank}(B) = n$, the coefficient matrix of the symplectic form ω is the inverse of the Poisson matrix, i.e. $P = B^{-1}$. In this setting, the cluster map is both a Poisson and a symplectic map. On the other hand, in the degenerate case $\text{rank}(B) < n$, the map is Poisson but not symplectic. A Poisson manifold equipped with a degenerate Poisson bracket admits a foliation by symplectic leaves, which enables one to study the dynamical system on each leaf, where the symplectic structure is non-degenerate. In this context, there is a canonical way to reduce a Poisson map to a symplectic map via the symplectic form. The following result is stated as Theorem 2.6 in [6].

Theorem 2.3 (Symplectic reduction) *In the degenerate case, rank $B = 2r < N$, there exists a rational map π ,*

$$\begin{aligned} \pi : \mathbb{C}^N &\rightarrow \mathbb{C}^{2r} \\ \mathbf{x} = (x_1, \dots, x_N) &\mapsto \mathbf{y} = (y_1, \dots, y_{2r}) \end{aligned} \quad (18)$$

equivalent to

$$y_j = \mathbf{x}^{\mathbf{v}^{(j)}} = \prod_i x_i^{v_i^{(j)}}, \quad \mathbf{v}^{(j)} \in \text{Im } B \quad (19)$$

which reduces the φ to the symplectic map $\hat{\varphi}$ satisfying the following relation,

$$\pi \cdot \varphi = \hat{\varphi} \cdot \pi$$

depicted by

$$\begin{array}{ccc} \mathbb{C}^N & \xrightarrow{\varphi} & \mathbb{C}^N \\ \downarrow \pi & & \downarrow \pi \\ \mathbb{C}^{2r} & \xrightarrow{\hat{\varphi}} & \mathbb{C}^{2r} \end{array}$$

Let M be a matrix whose first $2r$ rows are $\mathbf{v}^{(i)} \in \text{Im } B$ for $1 \leq i \leq 2r$, and whose remaining $N - 2r$ rows are $\mathbf{u}^{(j)} \in \ker B$ for $2r + 1 \leq j \leq N$. The symplectic form $\hat{\omega}$ associated with $\hat{\varphi}$ is expressed in the following form,

$$\hat{\omega} = \sum_{i < j} \frac{\hat{b}_{ij}}{y_i y_j} dy_i \wedge dy_j. \quad (20)$$

The coefficients $\hat{b}_{ij}$ are entries of the new exchange matrix which satisfies

$$M^{-T} B M^{-1} = \begin{pmatrix} \hat{B} & 0 \\ 0 & 0 \end{pmatrix}, \quad (21)$$

and the pull-back relation $\pi^* \hat{\omega} = \omega$ holds.

3 Deformation of Integrable Cluster Map Associated with Type D_4

In this section, we revisit one of the results in [12], that is, construction of the two parameter family of integrable map on the plane, which is obtained by deformed mutations in the cluster algebra of type D_4 .

3.1 Periodic Cluster Map Associated with Type D_4

Let us consider Cartan matrix of Dynkin type D_4 ,

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix} \quad (22)$$

The exchange matrix corresponding to the Cartan matrix above is given by

$$B = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad (23)$$

Starting with setting the initial seed $(\mathbf{x}, B)$, consisting of the initial cluster $\mathbf{x} = (x_1, x_2, x_3, x_4)$ and the exchange matrix, we consider a specific sequence of mutations in the cluster algebra $\mathcal{A}(\mathbf{x}, B)$. One can verify that the exchange matrix is mutation periodic under this specific composition of matrix mutations, that is to say,

$$\mu_4 \mu_3 \mu_2 \mu_1(B) = B \quad (24)$$

Under the same sequence of mutations, the seed goes through the following transformations,

$$\begin{aligned} (x_1, x_2, x_3, x_4) &\xrightarrow{\mu_1} (x'_1, x_2, x_3, x_4) \xrightarrow{\mu_2} (x'_1, x'_2, x_3, x_4) \xrightarrow{\mu_3} (x'_1, x'_2, x'_3, x_4) \\ &\xrightarrow{\mu_4} (x'_1, x'_2, x'_3, x'_4) \end{aligned}$$

where each variable x_i is replaced by a new cluster variable x'_i in each mutations which is given by

$$\begin{aligned} \mu_1 : (x_1, x_2, x_3, x_4) &\mapsto (x'_1, x_2, x_3, x_4) : x'_1 x_1 = 1 + x_2 \\ \mu_2 : (x'_1, x_2, x_3, x_4) &\mapsto (x'_1, x'_2, x_3, x_4) : x'_2 x_2 = 1 + x'_1 x_3 x_4 \\ \mu_3 : (x'_1, x'_2, x_3, x_4) &\mapsto (x'_1, x'_2, x'_3, x_4) : x'_3 x_3 = 1 + x'_2 \\ \mu_4 : (x'_1, x'_2, x'_3, x_4) &\mapsto (x'_1, x'_2, x'_3, x'_4) : x'_4 x_4 = 1 + x'_2. \end{aligned} \quad (25)$$

From these facts above, one can construct the corresponding cluster map $\varphi_{D_4} = \mu_4 \mu_3 \mu_2 \mu_1$

$$\varphi_{D_4} : \mathbf{x} = (x_1, x_2, x_3, x_4) \mapsto \mathbf{x}' = (x'_1, x'_2, x'_3, x'_4). \quad (26)$$

This map is defined by the exchange relations shown in (25). Recall that a specific sequence of mutations in a cluster algebra of finite type has the property of Zamolodchikov periodicity with period $(h+2)/2$, as stated in Theorem 2.2. Since the Coxeter number of type D_4 is 6, the cluster map is periodic with period 4,

$$\varphi_{D_4}^4(\mathbf{x}) = \mathbf{x}.$$

Furthermore it preserves the following presymplectic 2-form,

$$\omega = \frac{1}{x_1 x_2} dx_1 \wedge dx_2 + \frac{1}{x_2 x_3} dx_2 \wedge dx_3 + \frac{1}{x_2 x_4} dx_2 \wedge dx_4. \quad (27)$$

Note that the presymplectic form is degenerate, as the rank of the exchange matrix B is 2, which is not full rank, i.e. $\text{rank}(B) = 2 < 4$. Therefore, we can perform a

symplectic reduction (see Theorem 2.3), which reduces the map to a symplectic map defined on the plane. We begin with considering the null space, $\ker B$, and the image, $\operatorname{im} B$, given by

$$\ker B = \langle (1, 0, 0, 1)^T, (1, 0, 1, 0)^T \rangle, \quad \operatorname{im} B = \langle (0, 1, 0, 0)^T, (-1, 0, 1, 1)^T \rangle. \quad (28)$$

Using the basis of $\operatorname{im} B$, we define a rational map $\pi : \mathbb{C}^4 \rightarrow \mathbb{C}^2$,

$$\begin{aligned} \pi : \quad & \mathbb{C}^4 \rightarrow \mathbb{C}^2 \\ \mathbf{x} = (x_1, x_2, x_3, x_4) & \mapsto \mathbf{u} = (u_1, u_2) \end{aligned} \quad (29)$$

which projects $\mathbf{x}$ onto the plane of reduced coordinates,

$$u_1 = x_2, \quad u_2 = \frac{x_3 x_4}{x_1} \quad (30)$$

Through this reduction π , the map φ_{D_4} reduces to the 2D symplectic map $\hat{\varphi}_{D_4}$, given by

$$\begin{aligned} \hat{\varphi}_{D_4} : \quad & \mathbb{C}^2 \rightarrow \mathbb{C}^2 \\ \mathbf{y} = (u_1, u_2) & \mapsto \left(\frac{1 + u_2 + u_1 u_2}{u_1}, \frac{(1 + u_1 + u_2 + u_1 u_2)^2}{u_2 u_1^2 (1 + u_1)} \right), \end{aligned} \quad (31)$$

which satisfies $\hat{\varphi} \cdot \pi = \pi \cdot \varphi$ and preserves the symplectic form,

$$\hat{\omega} = \frac{1}{u_1 u_2} du_1 \wedge du_2, \quad \pi^*(\hat{\omega}) = \omega. \quad (32)$$

To show that $\hat{\varphi}_{D_4}$ is Liouville integrable, it is sufficient to find a single function (a first integral) that is invariant under the action of the map. Earlier in this section, we pointed out that the cluster map φ_{D_4} is periodic with period 4. Consequently, $\hat{\varphi}_{D_4}$ is also periodic, since it is induced from φ_{D_4} via the projection π . Using this periodicity, one can construct the following symmetric function.

$$\begin{aligned} I(\mathbf{u}) &= \sum_{i=0}^3 (\hat{\varphi}_{D_4}^*)^i(u_1) \\ &= u_1 + u_2 + \frac{u_1}{u_2} + \frac{u_2}{u_1} + \frac{1}{u_2} + \frac{1}{u_1} + \frac{u_2}{u_1^2} + \frac{1}{u_2 u_1} + \frac{1}{u_1^2} + \frac{1}{u_2 u_1^2} \end{aligned} \quad (33)$$

which satisfies $\hat{\varphi}_{D_4}^*(I(\mathbf{u})) = I(\mathbf{u})$.

Remark 3.1 The reason we prefer working on the reduced space is that there are examples which are not integrable in the original space on which they are defined, but turn out to be integrable on the reduced space. For further details please see [6, 10]

In the next section, we adjust the cluster mutations (a deformation introduced in [11]) in (25) and find the conditions under which the map remains to be integrable.

3.2 Deformed Map of type D_4

We make a minor modification of the cluster mutations in the cluster map φ_{D_4} in a way that an arbitrary coefficient is attached to each monomial on the right hand side of expressions in (25) as shown below.

$$\begin{aligned}\tilde{\mu}_1 : (x_1, x_2, x_3, x_4) &\mapsto (x'_1, x_2, x_3, x_4) : x'_1 x_1 = b_1 + a_1 x_2 \\ \tilde{\mu}_2 : (x'_1, x_2, x_3, x_4) &\mapsto (x'_1, x'_2, x_3, x_4) : x'_2 x_2 = b_2 + a_2 x'_1 x_3 x_4 \\ \tilde{\mu}_3 : (x'_1, x'_2, x_3, x_4) &\mapsto (x'_1, x'_2, x'_3, x_4) : x'_3 x_3 = b_3 + a_3 x'_2 \\ \tilde{\mu}_4 : (x'_1, x'_2, x'_3, x_4) &\mapsto (x'_1, x'_2, x'_3, x'_4) : x'_4 x_4 = b_4 + a_4 x'_2\end{aligned}\quad (34)$$

which we refer to as the deformed map of type D_4 , $\phi_{D_4} = \tilde{\mu}_4 \tilde{\mu}_3 \tilde{\mu}_2 \tilde{\mu}_1$. We can reduce the number of parameters by scaling each cluster variable $x_i \rightarrow \lambda_i x_i$ with an appropriate choice of parameters $(\lambda_1, \lambda_2, \lambda_3, \lambda_4) \in (\mathbb{C}^*)^4$, which simplifies the process of calculation at a later stage. The iteration of the deformed map can be written as

$$\begin{aligned}x_{1,n+1} x_{1,n} &= b_1 + x_{2,n} \\ x_{2,n+1} x_{2,n} &= b_2 + x_{1,n+1} x_{3,n} x_{4,n} \\ x_{3,n+1} x_{3,n} &= b_3 + x_{2,n+1} \\ x_{4,n+1} x_{4,n} &= b_4 + x'_{2,n+1}\end{aligned}\quad (35)$$

where n denotes the number of iterates of $\tilde{\phi}$ i.e. $\tilde{\phi}^n(\mathbf{x})$. By Theorem 1.3 in [11]. Following the same steps as previously, we can project this map to a 2D parametric birational map via the rational map $\pi : \mathbb{C}^4 \rightarrow \mathbb{C}^2$ as

$$\begin{aligned}\hat{\phi}_{D_4} : \mathbb{C}^2 &\rightarrow \mathbb{C}^2 \\ \mathbf{u} = (u_1, u_2) &\mapsto \left(\frac{(b_1 + u_1)u_2 + b_2}{u_1}, \right. \\ &\quad \left. \frac{(b_4 + u_2)u_1 + b_1 u_2 + b_2}{u_2 u_1^2 (b_1 + u_1)} \right),\end{aligned}\quad (36)$$

which is intertwined with ϕ_{D_4} via π , i.e. $\hat{\phi}_{D_4} \cdot \pi = \pi \cdot \phi_{D_4}$, and preserves the symplectic form (32). To verify the Liouville integrability of the map, we first assume that corresponding first integral takes a form analogous to (33) such that

$$\tilde{I}(\mathbf{u}) = \alpha_0 u_1 + \alpha_1 u_2 + \frac{\alpha_2 u_1}{u_2} + \frac{\alpha_3 u_2}{u_1} + \frac{\alpha_4}{u_2} + \frac{\alpha_5}{u_1} + \frac{\alpha_6 u_2}{u_1^2} + \frac{\alpha_7}{u_2 u_1} + \frac{\alpha_8}{u_1^2} + \frac{\alpha_9}{u_2 u_1^2} \quad (37)$$

where we adjusted each monomial in the first integral (33) by attaching an arbitrary parameter α_i . Without loss of generality, we can set $\alpha_0 = 1$. Imposing invariance of $\tilde{I}$ under the action of $\hat{\phi}_{D_4}$, i.e. $\hat{\phi}_{D_4}^*(\tilde{I}(\mathbf{u})) = \tilde{I}(\mathbf{u})$, constrains parameters, yielding

$$\tilde{I}(\mathbf{u}) = u_1 + u_2 + \frac{u_1}{u_2} + \frac{(b_1 + 1)u_2}{u_1} + \frac{b_3 + b_4 + 1}{u_2} + \frac{b_1 + b_2 + b_3 + b_4 + 1}{u_1}$$

$$+ \frac{b_1 u_2}{u_1^2} + \frac{b_3 b_4 + b_3 + b_4}{u_1 u_2} + \frac{2b_2}{u_1^2} + \frac{b_3 b_4}{u_1^2 u_2}, \quad (38)$$

provided that one of the following sets of conditions is satisfied,

- (1) $b_2 = b_4 = b_1 b_3$
- (2) $b_1 = b_2 = b_3 b_4$
- (3) $b_2 = b_3 = b_1 b_4$

This result shows that the cluster map φ_{D_4} admits deformations into integrable maps in more than one distinct way. Notice that both the first integral and the deformed map are invariant under the exchange of parameters $b_3 \leftrightarrow b_4$ and original variables $x_3 \leftrightarrow x_4$. As a result, among the three cases considered, integrable maps corresponding to cases (1) and (2) are genuinely distinct.

$$\hat{\phi}_1 : \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \mapsto \left(\frac{(b_1 + u_1)u_2 + b_1 b_3}{u_1}, \frac{[(b_1 + u_1)u_2 + b_1 b_3(u_1 + 1)] \cdot [u_2 + b_3]}{u_1^2 u_2} \right) \quad (39)$$

$$\hat{\phi}_2 : \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \mapsto \left(\frac{(b_3 b_4 + u_1)u_2 + b_3 b_4}{u_1}, \frac{[(b_4 + u_2)u_1 + b_3 b_4(u_2 + 1)] \cdot [(b_3 + u_2)u_1 + b_3 b_4(u_2 + 1)]}{u_1^2 u_2 (b_3 b_4 + u_1)} \right) \quad (40)$$

Thus, the deformed map preserves both integrability and symplectic structure after deformation, but in the process, it loses the Laurent property, which is an essential feature of cluster algebras. In our previous work [12], we showed that each map admits *Laurentification* that lifts each deformed map to a cluster algebra of rank 8 (including 2 frozen variables), $\mathcal{A}(\tilde{\mathbf{x}}, Q_{D_4}^{(l)})$ endowed with cluster $\tilde{\mathbf{x}} = (\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_{10})$ and the extended quiver $Q_{D_4}^{(l)}$, for $l = \{1, 2\}$ (shown in Fig. 6).

The process begins with defining the rational map $\pi_1 : (\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_8) \rightarrow (u_1, u_2)$, which is equivalent to

$$u_1 = \frac{\tilde{x}_8}{\tilde{x}_5 \tilde{x}_6}, \quad u_2 = \frac{\tilde{x}_1 \tilde{x}_2 \tilde{x}_4}{\tilde{x}_3 \tilde{x}_6 \tilde{x}_7}, \quad (41)$$

and whose structure is derived from analysing the singularity confinement pattern using an empirical version of p -adic analysis (see further details in [12]). Using this rational map π_1 , one can pullback the symplectic form $\hat{\omega}$ to define a new symplectic form

$$\tilde{\omega} = \pi_1^*(\hat{\omega}) = \sum_{i < j} \frac{\tilde{b}_{ij}}{\tilde{x}_i \tilde{x}_j} d\tilde{x}_i \wedge d\tilde{x}_j \quad (42)$$

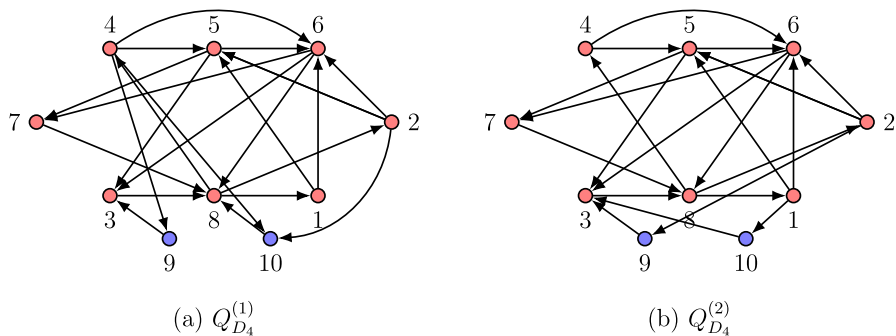


Fig. 6 (a) corresponds to $\hat{\phi}_1$, while (b) corresponds to $\hat{\phi}_2$

where the coefficients $\tilde{b}_{ij}$ denote the number of edges between mutable (non frozen) nodes in $Q_{D_4}^{(l)}$, directed from node i to node j . For each quiver, there exists a corresponding sequence of mutations that produces a cluster map. Each such cluster map arises from the deformed maps via Laurentification in the cluster algebra $\mathcal{A}(\tilde{\mathbf{x}}, Q_{D_4}^{(l)})$. The map corresponding to $Q_{D_4}^{(1)}$ is given by

$$\psi_{D_4}^{(1)} = \pi_1^* \hat{\phi}_{D_4}^{(1)} = \hat{\rho}_1^{-1} \hat{\mu}_2 \hat{\mu}_3 \hat{\mu}_8, \quad \rho_1 = (3, 1, 8)(2, 5, 6, 7, 4) \quad (43)$$

whose mutations are defined in cluster algebra $\mathcal{A}(\tilde{\mathbf{x}}, Q_{D_4}^{(1)})$. Note that here we denote the mutations in this cluster algebra $\hat{\mu}$ to distinguish them from the mutations in the cluster algebra of type D_4 . By setting the initial cluster $\tilde{\mathbf{x}}$ as

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3, \tilde{x}_4, \tilde{x}_5, \tilde{x}_6, \tilde{x}_7, \tilde{x}_8, \tilde{x}_9, \tilde{x}_{10}) = (\sigma_1, \tau_0, \sigma_0, \tau_4, \tau_1, \tau_2, \tau_3, r_1, b_3, b_4)$$

the action of the map $\psi_{D_4}^{(1)}$ induces

$$\begin{aligned} \psi_{D_4}^{(1)} : (\tilde{x}_1, \tilde{x}_2, \tilde{x}_3, \tilde{x}_4, \tilde{x}_5, \tilde{x}_6, \tilde{x}_7, \tilde{x}_8, \tilde{x}_9, \tilde{x}_{10}) &\rightarrow (\tilde{x}'_8, \tilde{x}_6, \tilde{x}_1, \tilde{x}'_2, \tilde{x}_7, \tilde{x}_5, \tilde{x}_4, \tilde{x}'_3, \tilde{x}_9, \tilde{x}_{10}) \\ (\sigma_1, \tau_0, \sigma_0, \tau_4, \tau_1, \tau_2, \tau_3, r_0, b_3, b_4) &\mapsto (\sigma_2, \tau_1, \sigma_1, \tau_5, \tau_2, \tau_3, \tau_4, r_1, b_3, b_4), \end{aligned} \quad (44)$$

The discrete evolution induced by the map is described by the following system of recurrence relations,

$$\begin{aligned} \sigma_{n+2} r_n &= b_3 \sigma_n \tau_{n+2} \tau_{n+3} + \sigma_{n+1} \tau_n \tau_{n+4} \\ r_{n+1} \sigma_n &= \sigma_{n+1} \tau_{n+4} \tau_n + b_1 \sigma_{n+2} \tau_{n+2} \tau_{n+1} \\ \tau_{n+5} \tau_n &= b_1 b_3 \tau_{n+2} \tau_{n+3} + r_{n+1}. \end{aligned} \quad (45)$$

Laurentification of the second deformed map $\hat{\phi}_2$ gives rises to following cluster map,

$$\psi_{D_4}^{(2)} = \pi_2^* \hat{\phi}_{D_4}^{(2)} = \hat{\rho}_2^{-1} \hat{\mu}_2 \hat{\mu}_1 \hat{\mu}_8 \hat{\mu}_3, \quad \hat{\rho}_2 = (1, 2)(3, 4, 5, 6, 7) \quad (46)$$

Now if we identify the cluster variables in the initial cluster $\tilde{\mathbf{x}}$ as

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3, \tilde{x}_4, \tilde{x}_5, \tilde{x}_6, \tilde{x}_7, \tilde{x}_8, \tilde{x}_9, \tilde{x}_{10}) = (\hat{s}_0, \hat{r}_0, \hat{t}_0, \hat{t}_1, \hat{t}_2, \hat{t}_3, \hat{t}_4, \hat{\eta}_0, b_3, b_4) \quad (47)$$

then we see that the iteration of the map,

$$\begin{aligned} \psi_{D_4}^{(1)} : (\tilde{x}_1, \tilde{x}_2, \tilde{x}_3, \tilde{x}_4, \tilde{x}_5, \tilde{x}_6, \tilde{x}_7, \tilde{x}_8, \tilde{x}_9, \tilde{x}_{10}) &\rightarrow (\tilde{x}'_2, \tilde{x}'_1, \tilde{x}'_3, \tilde{x}_4, \tilde{x}_5, \tilde{x}_6, \tilde{x}_7, \tilde{x}_3, \tilde{x}'_8, \tilde{x}_9, \tilde{x}_{10}) \\ (\hat{s}_0, \hat{r}_0, \hat{t}_0, \hat{t}_1, \hat{t}_2, \hat{t}_3, \hat{t}_4, \hat{\eta}_0, b_3, b_4) &\mapsto (\hat{s}_1, \hat{r}_1, \hat{t}_1, \hat{t}_2, \hat{t}_3, \hat{t}_4, \hat{t}_5, \hat{\eta}_1, b_3, b_4), \end{aligned} \quad (48)$$

is equivalent to the following recursion relations,

$$\begin{aligned} \hat{t}_{n+5} \hat{t}_n &= b_3 b_4 \hat{t}_{n+3} \hat{t}_{n+2} + \hat{\eta}_n, \\ \hat{\eta}_{n+1} \hat{\eta}_n &= \hat{r}_n \hat{\sigma}_n \hat{t}_{n+1} \hat{t}_{n+5} + b_3 b_4 \hat{t}_{n+4} \hat{t}_{n+3}^2 \hat{t}_{n+2}, \\ \hat{r}_{n+1} \hat{s}_n &= b_3 \hat{t}_{n+3} \hat{t}_{n+4} + \hat{\eta}_{n+1}, \\ \hat{s}_{n+1} \hat{r}_n &= b_4 \hat{t}_{n+3} \hat{t}_{n+4} + \hat{\eta}_{n+1}. \end{aligned} \quad (49)$$

In summary, the deformation approach introduced by [11] generalizes the Liouville integrable type D_4 cluster map in two distinct ways. Moreover, this approach yields new cluster algebras via Laurentification, with quivers that induce cluster maps, $\psi^{(1)}$ and $\psi^{(2)}$, corresponding to each deformed map, $\hat{\varphi}_1$ and $\hat{\varphi}_2$ respectively.

4 Deformed Cluster Map of Type D_6

The Cartan matrix for the type D_6 is

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & -1 & 0 & 2 \end{pmatrix} \quad (50)$$

which corresponds to the following exchange matrix,

$$B = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \end{pmatrix} \quad (51)$$

The associated cluster map φ_{D_6} is given by the composition of mutations $\mu_6 \mu_5 \mu_4 \mu_3 \mu_2 \mu_1$, which preserves the type D_6 quiver. Let us consider the deformation of each

mutation in the cluster map φ_{D_6} which is shown below.

$$\begin{aligned}
 \mu_1 : (x_1, x_2, x_3, x_4, x_5, x_6) &\mapsto (x'_1, x_2, x_3, x_4, x_5, x_6), & x_1 x'_1 &= b_1 + a_1 x_2 \\
 \mu_2 : (x'_1, x_2, x_3, x_4, x_5, x_6) &\mapsto (x'_1, x'_2, x_3, x_4, x_5, x_6), & x_2 x'_2 &= b_2 + a_2 x_3 x'_1 \\
 \mu_3 : (x'_1, x'_2, x_3, x_4, x_5, x_6) &\mapsto (x'_1, x'_2, x'_3, x_4, x_5, x_6), & x_3 x'_3 &= b_3 + a_3 x_4 x'_2 \\
 \mu_4 : (x'_1, x'_2, x'_3, x_4, x_5, x_6) &\mapsto (x'_1, x'_2, x'_3, x'_4, x_5, x_6), & x_4 x'_4 &= b_4 + a_4 x_5 x_6 x'_3 \\
 \mu_5 : (x'_1, x'_2, x'_3, x'_4, x_5, x_6) &\mapsto (x'_1, x'_2, x'_3, x'_4, x'_5, x_6), & x_5 x'_5 &= b_5 + a_5 x'_4 \\
 \mu_6 : (x'_1, x'_2, x'_3, x'_4, x'_5, x_6) &\mapsto (x'_1, x'_2, x'_3, x'_4, x'_5, x'_6), & x_6 x'_6 &= b_6 + a_6 x'_4
 \end{aligned} \tag{52}$$

The deformed map $\tilde{\varphi}_{D_6} = \tilde{\mu}_6 \tilde{\mu}_5 \tilde{\mu}_4 \tilde{\mu}_3 \tilde{\mu}_2 \tilde{\mu}_1$ transforms back to the original cluster map $\varphi_{D_6} = \mu_6 \mu_5 \mu_4 \mu_3 \mu_2 \mu_1$ when we fix the parameters $a_i = 1 = b_i$ for all $i = 1, \dots, 4$. Furthermore, since the Coxeter number h for the type D_6 is 10, the periodicity for the cluster map is period 6,

$$\varphi_{D_6} \cdot (\mathbf{x}, B) = (\varphi_{D_6}(\mathbf{x}), B) \text{ and } \varphi^6(\mathbf{x}) = \mathbf{x} \tag{53}$$

The symplectic form ω , that is invariant under the deformed map, takes the following form,

$$\begin{aligned}
 \omega &= \frac{1}{x_1 x_2} dx_1 \wedge dx_2 + \frac{1}{x_2 x_3} dx_2 \wedge dx_3 \\
 &\quad + \frac{1}{x_4 x_5} dx_4 \wedge dx_5 + \frac{1}{x_4 x_6} dx_4 \wedge dx_6
 \end{aligned} \tag{54}$$

As in the previous example, we rescale each cluster variable (i.e. $x_i \rightarrow \lambda_i x_i$) so that the (52) can be rewritten as

$$\begin{aligned}
 x_{1,n+1} x_{1,n} &= x_{2,n} + b_1, \\
 x_{2,n+1} x_{2,n} &= x_{3,n} x_{1,n+1} + b_2, \\
 x_{3,n+1} x_{3,n} &= x_{4,n} x_{2,n+1} + b_3 \\
 x_{4,n+1} x_{4,n} &= x_{5,n} x_{6,n} x_{3,n+1} + b_4 \\
 x_{5,n+1} x_{5,n} &= x_{4,n+1} + b_5 \\
 x_{6,n+1} x_{6,n} &= x_{4,n+1} + b_6
 \end{aligned} \tag{55}$$

From the exchange matrix B_{D_6} , (51), one can see that it is degenerate and possesses rank 4. Thus we consider the null space and column space of B_{D_6} given by the following

$$\begin{aligned}
 \ker(B) &= \left\{ (1, 0, 1, 0, 0, 1)^T, (1, 0, 1, 0, 1, 0)^T \right\}, \\
 \operatorname{im}(B) &= \left\{ (0, 1, 0, 0, 0, 0)^T, (-1, 0, 1, 0, 0, 0)^T, (0, -1, 0, 1, 0, 0)^T, \right. \\
 &\quad \left. (0, 0, -1, 0, 1, 1)^T \right\}
 \end{aligned} \tag{56}$$

The vectors in $\operatorname{im}(B)$ lead to constructing new variables y_1, y_2, y_3, y_4 i.e.

$$y_1 = x_2, \quad y_2 = \frac{x_3}{x_1}, \quad y_3 = \frac{x_4}{x_2}, \quad y_4 = \frac{x_5 x_6}{x_3} \tag{57}$$

which reduces the deformed map to the 4D symplectic map i.e.

$$\hat{\varphi} : (y_1, y_2, y_3, y_4) \rightarrow (y'_1, y'_2, y'_3, y'_4) \quad (58)$$

where the produced variables are written as

$$\begin{aligned} y'_1 &= \frac{(b_1 + y_1)y_2 + b_2}{y_1} \\ y'_2 &= \frac{y_3(b_1 + y_1)y_2 + b_2y_3 + b_3}{y_2(b_1 + y_1)} \\ y'_3 &= \frac{y_3(b_1 + y_1)y_2y_4 + b_2y_3y_4 + b_3y_4 + b_4}{y_3(b_2 + (b_1 + y_1)y_2)} \\ y'_4 &= \frac{((b_1y_2 + y_1y_2 + b_2)y_3y_4 + b_6y_1y_3 + b_3y_4 + b_4)((b_1y_2 + y_1y_2 + b_2)y_3y_4 + b_5y_1y_3 + b_3y_4 + b_4)}{y_1^2y_3^2y_4((b_1y_2 + y_1y_2 + b_2)y_3 + b_3)} \end{aligned} \quad (59)$$

This reduced map preserves the nondegenerate symplectic form

$$\hat{\omega} = \sum_{ij} \hat{b}_{ij} d \log y_i \wedge d \log y_j$$

whose coefficients are entries of the following 4×4 skew-symmetric matrix

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & -1 & 0 \end{pmatrix}$$

4.1 Laurentification of Deformed Type D_6 map

From the several successful cases shown in [7, 11, 12] (including the previous section), we observed that the degrees (or magnitude) of numerators and denominators in the iterates grow exponentially when arbitrary parameter values are used. However when the integrability condition is imposed, this growth is reduced significantly. This hints that choosing suitable conditions on the parameters leads to cancellation between the numerator and denominator, simplifying the rational expressions and hence reducing the complexity of the map. This reduced growth corresponds to the notion of degree growth used in *algebraic entropy*, which is an algebraic test for integrability. Under these conditions, one was able to find the singularity confinement patterns which allow us to apply Laurentification and lift the map to a space where the Laurent property holds. By combining altogether, this indicates a close relation between slow growth (slower than exponential growth) and Laurentification.

We begin by considering the iteration of the deformed type D_6 map, with the parameters chosen so that $b_1 = b_2 = b_3 = b_4 = b_5b_6$. We study the heights $H(\mathbf{y}_n)$ of the iterates $\mathbf{y}_n = (y_{j,n})_{1 \leq j \leq 4}$ of the deformed map $\hat{\varphi}_{D_6}$, which measure the complexity growth of the map numerically. The height is defined by

$$H(\mathbf{y}_n) = \max_{1 \leq j \leq 4} \max(|u_{j,n}|, |v_{j,n}|)$$

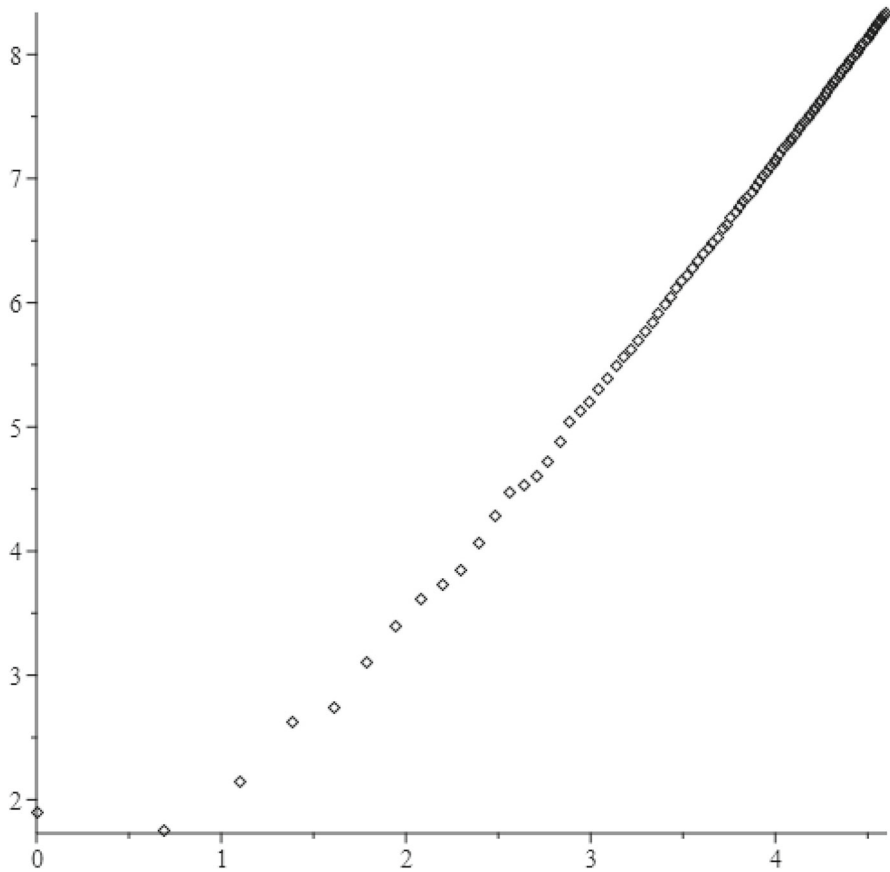


Fig. 7 x -axis represents $\log n$ and y -axis corresponds to $\log H(\mathbf{y}_n)$

where $u_{j,n}$ and $v_{j,n}$ are the numerator and denominator of the iterates $y_{j,n}$, and they are coprime. A standard way to test for subexponential growth is to compare the logarithmic height $\log H(\mathbf{y}_n)$ with $\log n$: if $\log H(\mathbf{y}_n)$ is asymptotically proportional to $\log n$, then $H(\mathbf{y}_n)$ grows polynomially. Setting initial values $y_1 = y_2 = y_3 = y_4 = 1$ and parameters $b_5 = 2$, $b_6 = 3$, we plot $\log H(\mathbf{y}_n)$ against $\log n$ which is illustrated in Fig. 7. From the graph, it is clear that the plot is asymptotically a straight line; the slope a of this line indicates that

$$\log H(\mathbf{y}_n) \sim a \log n \implies H(\mathbf{y}_n) \sim n^a$$

Thus the height exhibits polynomial growth n^a for some $a > 0$. This growth is therefore slower than exponential growth.

Next, using the same initial values and parameter values as mentioned earlier, we consider the sequence of rational values, which are factorized into prime numbers as shown in the table.

n	$y_{1,n}$	$y_{2,n}$	$y_{3,n}$	$y_{4,n}$
1	13	$\frac{19}{7}$	$\frac{5^2}{13}$	$\frac{2^2 \cdot 3^3 \cdot 7}{19}$
2	$\frac{31}{7}$	$\frac{43}{19}$	$2 \cdot 3 \cdot 7$	$\frac{7 \cdot 47}{43}$
3	$\frac{127}{19}$	$\frac{2^3 \cdot 3 \cdot 7 \cdot 23}{73}$	$\frac{977}{127}$	$\frac{5 \cdot 11 \cdot 29}{2^2 \cdot 3 \cdot 19 \cdot 23}$
4	$\frac{2 \cdot 3^3 \cdot 137}{73}$	$\frac{13109}{7 \cdot 241}$	$\frac{2 \cdot 23 \cdot 467}{3^3 \cdot 19 \cdot 137}$	$\frac{3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13}{73 \cdot 13109}$
5	$\frac{5 \cdot 2797}{7 \cdot 241}$	$\frac{47 \cdot 499}{2 \cdot 3 \cdot 19 \cdot 653}$	$\frac{3 \cdot 7 \cdot 36263}{5 \cdot 73 \cdot 2797}$	$\frac{2^4 \cdot 17 \cdot 19 \cdot 443}{47 \cdot 241 \cdot 499}$
6	$\frac{94307}{2 \cdot 3 \cdot 19 \cdot 653}$	$\frac{3 \cdot 7 \cdot 257501}{73 \cdot 24107}$	$\frac{2^2 \cdot 19 \cdot 23 \cdot 19259}{241 \cdot 94307}$	$\frac{73 \cdot 13457 \cdot 5197}{3^2 \cdot 653 \cdot 257501}$
7	$\frac{3 \cdot 13117691}{73 \cdot 24107}$	$\frac{2^4 \cdot 19 \cdot 211 \cdot 30559}{7 \cdot 241 \cdot 269 \cdot 2011}$	$\frac{5^2 \cdot 73 \cdot 2711 \cdot 62311}{3^2 \cdot 653 \cdot 13117691}$	$\frac{2^5 \cdot 7 \cdot 23 \cdot 41 \cdot 181 \cdot 241 \cdot 881}{211 \cdot 30559 \cdot 24107}$
8	$\frac{2 \cdot 199 \cdot 980249}{241 \cdot 269 \cdot 2011}$	$\frac{2^3 \cdot 13 \cdot 73 \cdot 149 \cdot 1053713}{3^3 \cdot 7 \cdot 11 \cdot 19 \cdot 653 \cdot 10289}$	$\frac{83 \cdot 229 \cdot 241 \cdot 6317 \cdot 8681}{199 \cdot 980249 \cdot 24107}$	$\frac{3^2 \cdot 11 \cdot 179 \cdot 653 \cdot 98597 \cdot 14281}{2^3 \cdot 13 \cdot 149 \cdot 269 \cdot 1053713 \cdot 2011}$
9	$\frac{19 \cdot 252533 \cdot 45127}{3^3 \cdot 7 \cdot 11 \cdot 653 \cdot 10289}$	$\frac{59 \cdot 97 \cdot 241 \cdot 605239213}{73 \cdot 7712933 \cdot 24107}$	$\frac{3^2 \cdot 11 \cdot 13 \cdot 109 \cdot 653 \cdot 74857 \cdot 31627}{19 \cdot 269 \cdot 252533 \cdot 45127 \cdot 2011}$	$\frac{5 \cdot 1543 \cdot 24107 \cdot 6089 \cdot 280759}{2 \cdot 3 \cdot 7 \cdot 59 \cdot 97 \cdot 605239213 \cdot 10289}$

To construct variable transformations which lift the deformed type D_6 map to a cluster algebra of higher rank, we study the singularity structure of the map via p-adic analysis, which is done by observing the patterns of prime factors appearing in the sequences. One can see that for each prime numbers $p_1 = 73, 241, 653$ (for instance), it appears in all variables y_i for $i = \{1, 2, 3, 4\}$. The corresponding p_1 -adic norm exhibits the following pattern

$$\begin{aligned}
 |y_{1,n}|_{p_1} &= 1, 1, p_1, 1, 1, p_1, 1 \\
 |y_{2,n}|_{p_1} &= 1, p_1, 1, 1, p_1, 1, p_1^{-1}, p_1, 1 \\
 |y_{3,n}|_{p_1} &= 1, 1, 1, p_1, 1, p_1^{-1}, 1, 1 \\
 |y_{4,n}|_{p_1} &= 1, 1, p_1, 1, p_1^{-1}, 1, 1
 \end{aligned} \tag{60}$$

There are particular values which emerge in two variables. For $p_2 = 31, 137$, one has $|y_{1,n}|_{p_2} = p_2^{-1}$ and $|y_{3,n}|_{p_2} = p_2$. For $p_3 = 19, 23, 43, 47, 499, 13109$, one has $|y_{2,n}|_{p_3} = p_3^{-1}$ and $|y_{4,n}|_{p_3} = p_3$. Each variable is associated with a set of primes that does not appear in the other variables; for instance, the primes $p = 31, 137$ arise exclusively in the y_1 . Thus one can observe the singularity confinement patterns in the iterations in $(y_{1,n}, y_{2,n}, y_{3,n}, y_{4,n})$

$$\begin{aligned}
 \text{Pattern 1 : } & \dots \rightarrow (R, \infty^1, R, R) \rightarrow (\infty^1, R, R, \infty^1) \rightarrow (R, R, \infty^1, R) \\
 & \rightarrow (R, \infty^1, R, 0^1) \rightarrow (\infty^1, R, 0^1, R) \rightarrow (R, 0^1, R, R) \\
 & \rightarrow (R, \infty^1, R, R) \rightarrow \dots \\
 \text{Pattern 2 : } & \dots \rightarrow (0^1, R, \infty, R) \rightarrow \dots \\
 \text{Pattern 3 : } & \dots \rightarrow (R, 0^1, R, \infty) \rightarrow \dots \\
 \text{Pattern 4 : } & \dots \rightarrow (R, R, 0^1, R) \rightarrow \dots \\
 \text{Pattern 5 : } & \dots \rightarrow (R, R, R, 0^1) \rightarrow \dots
 \end{aligned} \tag{61}$$

Given tau-functions specified by $\tau \equiv 0 \pmod{p_1}$, $\chi \equiv 0 \pmod{p_2}$, $\eta \equiv 0 \pmod{p_3}$, $\xi \equiv 0 \pmod{p_4}$, $r \equiv 0 \pmod{p_5}$, $s \equiv 0 \pmod{p_6}$, the symplectic coordinates can be written as

$$y_{1,n} = \frac{\chi_n}{\tau_{n+5}\tau_{n+2}}, \quad y_{2,n} = \frac{\tau_{n+1}\eta_{1,n}}{\tau_n\tau_{n+3}\tau_{n+6}}, \quad y_{3,n} = \frac{\tau_{n+2}\xi_{1,n}}{\tau_{n+4}\chi_n}, \quad y_{4,n} = \frac{\tau_{n+3}r_{n+1}}{\tau_{n+5}\eta_{1,n}} \quad (62)$$

so that the confinement patterns are reproduced under the iteration of the map. Substituting these variables into the exchange relations yields new expressions, but they are not in the form of an exchange relation. Instead, we consider the original system formulated with the x -variables. With the initial cluster variables $x_i = 1$ for $i = 1, 2, 3, 4, 5, 6$ and taking the same parameter values as above, the iteration then produces the sequences shown in the table below.

n	$x_{1,n}$	$x_{2,n}$	$x_{3,n}$	$x_{4,n}$
1	7	13	19	5^2
2	$\frac{19}{7}$	$\frac{31}{7}$	$\frac{43}{7}$	$2 \cdot 3 \cdot 31$
3	$\frac{73}{19}$	$\frac{127}{19}$	$\frac{2^3 \cdot 3 \cdot 7 \cdot 23}{19}$	$\frac{977}{19}$
4	$\frac{241}{73}$	$\frac{2 \cdot 3^3 \cdot 137}{73}$	$\frac{13109}{7 \cdot 73}$	$\frac{2^2 \cdot 23 \cdot 467}{19 \cdot 73}$
5	$\frac{2^2 \cdot 3 \cdot 653}{241}$	$\frac{5 \cdot 2797}{7 \cdot 241}$	$\frac{2 \cdot 47 \cdot 499}{19 \cdot 241}$	$\frac{3 \cdot 36263}{73 \cdot 241}$
6	$\frac{24107}{2^2 \cdot 3 \cdot 7 \cdot 653}$	$\frac{94507}{2 \cdot 3 \cdot 19 \cdot 653}$	$\frac{257501}{2^2 \cdot 73 \cdot 653}$	$\frac{2 \cdot 25 \cdot 19259}{3 \cdot 241 \cdot 653}$
7	$\frac{2 \cdot 7 \cdot 269 \cdot 2011}{19 \cdot 24107}$	$\frac{3 \cdot 13117691}{73 \cdot 24107}$	$\frac{2^3 \cdot 211 \cdot 30559}{241 \cdot 24107}$	$\frac{5^2 \cdot 2711 \cdot 62311}{3 \cdot 653 \cdot 24107}$
8	$\frac{3^2 \cdot 7 \cdot 11 \cdot 19 \cdot 10289}{2 \cdot 73 \cdot 269 \cdot 2011}$	$\frac{2 \cdot 199 \cdot 980249}{241 \cdot 269 \cdot 2011}$	$\frac{2^2 \cdot 13 \cdot 149 \cdot 1053713}{3 \cdot 269 \cdot 653 \cdot 2011}$	$\frac{2 \cdot 83 \cdot 229 \cdot 6317 \cdot 8681}{269 \cdot 24107 \cdot 2011}$

n	$x_{5,n}$	$x_{6,n}$
1	3^3	$2^2 \cdot 7$
2	$\frac{2^2 \cdot 47}{3^3}$	$\frac{3^3}{2^2}$
3	$\frac{3^3 \cdot 5 \cdot 7 \cdot 29}{2^2 \cdot 19 \cdot 47}$	$\frac{2^3 \cdot 11 \cdot 47}{3^3 \cdot 19}$
4	$\frac{2^3 \cdot 11^2 \cdot 47}{5 \cdot 29 \cdot 73}$	$\frac{3^3 \cdot 5^3 \cdot 13 \cdot 29}{2^3 \cdot 11 \cdot 47 \cdot 73}$
5	$\frac{5^3 \cdot 13 \cdot 29 \cdot 443}{2^3 \cdot 11^2 \cdot 47 \cdot 241}$	$\frac{2^8 \cdot 11^2 \cdot 17 \cdot 47}{5^3 \cdot 13 \cdot 29 \cdot 241}$
6	$\frac{26 \cdot 11^2 \cdot 17 \cdot 47 \cdot 13457}{3 \cdot 5^3 \cdot 13 \cdot 29 \cdot 443 \cdot 653}$	$\frac{5^3 \cdot 13 \cdot 29 \cdot 443 \cdot 5197}{2^8 \cdot 3 \cdot 11^2 \cdot 17 \cdot 47 \cdot 653}$
7	$\frac{5^3 \cdot 13 \cdot 23 \cdot 29 \cdot 41 \cdot 443 \cdot 881 \cdot 5197}{2^9 \cdot 11^2 \cdot 17 \cdot 47 \cdot 13457 \cdot 24107}$	$\frac{2^1 \cdot 6 \cdot 7 \cdot 11^2 \cdot 17 \cdot 47 \cdot 181 \cdot 13457}{5^3 \cdot 13 \cdot 29 \cdot 443 \cdot 24107 \cdot 5197}$
8	$\frac{2^{15} \cdot 3 \cdot 7 \cdot 11^3 \cdot 17 \cdot 47 \cdot 181 \cdot 98597 \cdot 13457}{5^3 \cdot 13 \cdot 23 \cdot 29 \cdot 41 \cdot 269 \cdot 443 \cdot 881 \cdot 5197 \cdot 2011}$	$\frac{5^3 \cdot 13 \cdot 23 \cdot 29 \cdot 41 \cdot 179 \cdot 443 \cdot 881 \cdot 5197 \cdot 14281}{2^1 \cdot 6 \cdot 7 \cdot 11^2 \cdot 17 \cdot 47 \cdot 181 \cdot 269 \cdot 13457 \cdot 2011}$

The prime numbers appearing in the y variables, once again can be spotted in the x -variables. e.g $p = 73, 241, 653$ corresponds to τ , $p = 31, 137$ associated with P , and $p = 19, 23, 43, 47, 499$ corresponds to Q . $p = 467, 977, 36263$ corresponds to R . There are specific primes in x_5 and x_6 which cancel out in $y_4 = x_5 x_6 / x_3$, for instance $p = 19, 73$. Following the same approach as previously, we define the new

tau function σ_n which is associated with such primes. Thus we can then construct explicit expressions for the tau-functions,

$$\begin{aligned} x_{1,n} &= \frac{\tau_{n+6}\tau_n}{\tau_{n+5}\tau_{n+1}}, & x_{2,n} &= \frac{\chi_n}{\tau_{n+5}\tau_{n+2}}, & x_{3,n} &= \frac{\eta_{1,n}}{\tau_{n+5}\tau_{n+3}} \\ x_{4,n} &= \frac{\xi_{1,n}}{\tau_{n+5}\tau_{n+4}}, & x_{5,n} &= \frac{r_n\sigma_n}{\tau_{n+5}}, & x_{6,n} &= \frac{s_n}{\tau_{n+5}\sigma_n} \end{aligned} \quad (63)$$

where $\sigma_n\sigma_{n+1} = \frac{s_n}{r_n}$. Then by directly substituting these variables into the deformed mutations, we obtain the following recurrence relations,

$$\begin{aligned} \tau_{n+7}\tau_n &= b_5b_6\tau_{n+5}\tau_{n+2} + \chi_n \\ \chi_{n+1}\chi_n &= b_5b_6\tau_{n+2}\tau_{n+3}\tau_{n+5}\tau_{n+6} + \eta_{1,n}\tau_{n+1}\tau_{n+7} \\ \eta_{1,n+1}\eta_{1,n} &= b_5b_6\tau_{n+3}\tau_{n+4}\tau_{n+5}\tau_{n+6} + \xi_{1,n}\chi_{n+1} \\ \xi_{1,n+1}\xi_{1,n} &= b_5b_6\tau_{n+4}\tau_{n+5}^2\tau_{n+6} + r_ns_n\eta_{1,n+1} \\ r_{n+1}s_n &= b_5\tau_{n+5}\tau_{n+6} + \xi_{1,n+1} \\ s_{n+1}r_n &= b_6\tau_{n+5}\tau_{n+6} + \xi_{1,n+1} \end{aligned} \quad (64)$$

which we denote as iterations of the corresponding birational map

$$\begin{aligned} \psi_{D_6} : & (\xi_{1,0}, \tau_0, \tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6, \chi_0, \eta_{1,0}, b_5, b_6) \\ \rightarrow & (\xi_{1,1}, \tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6, \tau_7, \chi_1, \eta_{1,1}, b_5, b_6) \end{aligned} \quad (65)$$

on the following initial data,

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_4, \tilde{x}_5, \tilde{x}_6, \tilde{x}_7, \tilde{x}_8, \tilde{x}_9, \tilde{x}_{10}, \tilde{x}_{11}, \tilde{x}_{12}) = (\xi_{1,0}, \tau_0, \tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6, \chi_0, \eta_{1,0}).$$

Let us denote the rational map $\pi : \mathbb{C}^{12} \rightarrow \mathbb{C}^4$ given by the variable transformations (63). Applying the pullback of the presymplectic form ω via the rational map π , yields a new presymplectic form whose coefficients give rise to a new exchange matrix, which is essential in defining ψ_{D_6} . Furthermore, we insert extra rows, $(0, 0, -1, 1, 0, 0, 0, 0, 0, 0, 0)^T$ and $(0, -1, 0, 1, 0, 0, 0, 0, 0, 0, 0)^T$ at the bottom of the matrix, thereby constructing the extended exchange matrix $\tilde{B}_{D_6}$ (67), which represents the quiver illustrated in Fig. 8.

As a result of these insertions, the parameters b_5 and b_6 appear in the cluster variables generated by mutations in the cluster algebra determined by the pair consisting of matrix $\tilde{B}_{D_6}$ (or, equivalently, the quiver Q_{D_6}) and initial cluster

$$\begin{aligned} \hat{\mathbf{x}} &= (\xi_{1,0}, r_0, s_0, \tau_0, \tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6, \chi_0, \eta_{1,0}, b_5, b_6) \\ &= (\tilde{x}_1, \tilde{x}_2, \tilde{x}_4, \tilde{x}_5, \tilde{x}_6, \tilde{x}_7, \tilde{x}_8, \tilde{x}_9, \tilde{x}_{10}, \tilde{x}_{11}, \tilde{x}_{12}, \tilde{x}_{13}, \tilde{x}_{14}). \end{aligned}$$

Thus we reach to the following result.

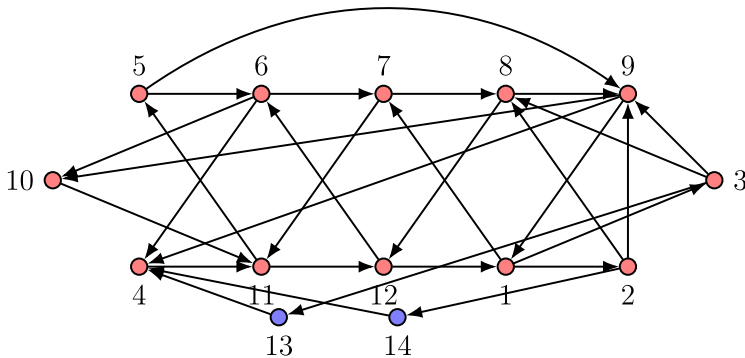


Fig. 8 Extended quiver Q_{D_6} associated with the deformed D_6

Theorem 4.1 (Laurentification of the deformed map) *Let $(\hat{\mathbf{x}}, \hat{B}_{D_6})$ be an initial seed which is composed of the extended initial cluster*

$$\begin{aligned}\hat{\mathbf{x}} &= (\xi_{1,0}, r_0, s_0, \tau_0, \tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6, \chi_0, \eta_{1,0}, b_5, b_6) \\ &= (\tilde{x}_1, \tilde{x}_2, \tilde{x}_4, \tilde{x}_5, \tilde{x}_6, \tilde{x}_7, \tilde{x}_8, \tilde{x}_9, \tilde{x}_{10}, \tilde{x}_{11}, \tilde{x}_{12}, \tilde{x}_{13}, \tilde{x}_{14})\end{aligned}\quad (66)$$

together with the associated extended exchange matrix

$$\tilde{B}_{D_6} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 1 & 0 & 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 1 \\ 1 & -1 & -1 & 1 & -1 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & -1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}\quad (67)$$

and consider the permutation $\rho = (45678910)(23)$. Then the iteration of the cluster map $\psi_{D_6} = \rho^{-1}\mu_{12}\mu_{11}\mu_{10}\mu_9\mu_8\mu_1$ is equivalent to the recurrence (64), and the tau functions $\tau_n, \chi_n, \eta_{1,n}, \xi_{1,n}, r_n, s_n$ are elements of the Laurent polynomial ring

$$\mathbb{Z}_{>0} \left[b_5, b_6, \chi_0^\pm, \eta_{1,0}^\pm, \xi_{1,0}^\pm, r_0^\pm, s_0^\pm, \tau_0^\pm, \tau_1^\pm, \tau_2^\pm, \tau_3^\pm, \tau_4^\pm, \tau_5^\pm, \tau_6^\pm \right].$$

4.2 Tropicalization and Degree Growth for Deformed D_6 map

In the previous work [12], we showed that the degree growth of the iteration of cluster maps $\psi_{D_4}^{(1)}$ and $\psi_{D_4}^{(2)}$, (induced from deformed type D_4 map introduced in Sect. 3.2) is quadratic, with a $(\max, +)$ relation corresponding to the exchange relations. This results in vanishing algebraic entropy, indicating that the maps are integrable, which is consistent with the integrability of the deformed map $\tilde{\varphi}_{D_4}$. In this section, we study the degree growth of cluster map ψ_{D_6} and determine associated algebraic entropy by following the same steps as in the type D_4 case [12].

As the Laurent property of the cluster map ψ_{D_6} holds, we can write the sequence of tau-functions τ_n , χ_n , $\eta_{1,n}$, $\xi_{1,n}$, r_n and s_n as Laurent polynomials in the initial cluster $\hat{\mathbf{x}}$ (66), shown below

$$\begin{aligned}\tau_n &= \frac{N_n^{(1)}(\hat{\mathbf{x}})}{\hat{\mathbf{x}}^{\mathbf{d}_n}}, & \chi_n &= \frac{N_n^{(2)}(\hat{\mathbf{x}})}{\hat{\mathbf{x}}^{\mathbf{p}_n}}, & \eta_{1,n} &= \frac{N_n^{(3)}(\hat{\mathbf{x}})}{\hat{\mathbf{x}}^{\mathbf{q}_n}}, \\ \xi_{1,n} &= \frac{N_n^{(4)}(\hat{\mathbf{x}})}{\hat{\mathbf{x}}^{\mathbf{r}_n}}, & r_n &= \frac{N_n^{(4)}(\hat{\mathbf{x}})}{\hat{\mathbf{x}}^{\mathbf{v}_n}}, & s_n &= \frac{N_n^{(4)}(\hat{\mathbf{x}})}{\hat{\mathbf{x}}^{\mathbf{j}_n}}\end{aligned}\quad (68)$$

where d-vectors (denominator vectors) $\mathbf{d}_n$, $\mathbf{p}_n$, $\mathbf{q}_n$, $\mathbf{r}_n$, $\mathbf{v}_n$, $\mathbf{j}_n$ are associated with the tau-functions associated with unfrozen variables ($\xi_{1,0}$, r_0 , s_0 , τ_0 , τ_1 , τ_2 , τ_3 , τ_4 , τ_5 , τ_6 , χ_0 , $\eta_{1,0}$, b_5 , b_6) and have initial data given by a 12×12 identity matrix.

$$(\mathbf{q}_0 \ \mathbf{r}_0 \ \mathbf{v}_0 \ \mathbf{j}_0 \ \mathbf{d}_0 \ \mathbf{d}_1 \ \mathbf{d}_2 \ \mathbf{d}_3 \ \mathbf{d}_4 \ \mathbf{d}_5 \ \mathbf{d}_6 \ \mathbf{p}_0) = -I \quad (69)$$

Direct substitution (68) into (64) and comparison of the exponents in the denominators on both sides yields the $(\max, +)$ relations for the d-vectors,

$$\begin{aligned}\mathbf{d}_{n+7} + \mathbf{d}_n &= \max(\mathbf{d}_{n+5} + \mathbf{d}_{n+2}, \mathbf{p}_n), \\ \mathbf{p}_{n+1} + \mathbf{p}_n &= \max(\mathbf{d}_{n+2} + \mathbf{d}_{n+3} + \mathbf{d}_{n+5} + \mathbf{d}_{n+6}, \mathbf{q}_n + \mathbf{d}_{n+1} + \mathbf{d}_{n+7}), \\ \mathbf{q}_{n+1} + \mathbf{q}_n &= \max(\mathbf{d}_{n+3} + \mathbf{d}_{n+4} + \mathbf{d}_{n+5} + \mathbf{d}_{n+6}, \mathbf{r}_n + \mathbf{p}_{n+1}), \\ \mathbf{r}_{n+1} + \mathbf{r}_n &= \max(\mathbf{d}_{n+4} + 2\mathbf{d}_{n+5} + \mathbf{d}_{n+6}, \mathbf{v}_n + \mathbf{j}_n + \mathbf{q}_{n+1}), \\ \mathbf{v}_{n+1} + \mathbf{j}_n &= \max(\mathbf{d}_{n+5} + \mathbf{d}_{n+6}, \mathbf{r}_{n+1}), \\ \mathbf{j}_{n+1} + \mathbf{v}_n &= \max(\mathbf{d}_{n+5} + \mathbf{d}_{n+6}, \mathbf{r}_{n+1}).\end{aligned}\quad (70)$$

Next we introduce quantities which is analogous to the tropical version of (63) as follows,

$$\begin{aligned}\mathbf{X}_{1,n} &= \mathbf{d}_n + \mathbf{d}_{n+6} - \mathbf{d}_{n+1} - \mathbf{d}_{n+5}, & \mathbf{X}_{2,n} &= \mathbf{p}_n - \mathbf{d}_{n+2} - \mathbf{d}_{n+5}, \\ \mathbf{X}_{3,n} &= \mathbf{q}_n - \mathbf{d}_{n+3} - \mathbf{d}_{n+5}, & \mathbf{X}_{4,n} &= \mathbf{r}_n - \mathbf{d}_{n+4} - \mathbf{d}_{n+5}, \\ \mathbf{X}_{5,n} &= \mathbf{v}_n + \mathbf{s}_n - \mathbf{d}_{n+5}, & \mathbf{X}_{6,n} &= \mathbf{j}_n - \mathbf{d}_{n+5} - \mathbf{s}_n,\end{aligned}\quad (71)$$

along with quantities corresponding to symplectic coordinates y_j (62), shown below.

$$\begin{aligned}\mathbf{Y}_{1,n} &= \mathbf{p}_n - \mathbf{d}_{n+2} - \mathbf{d}_{n+5}, & \mathbf{Y}_{2,n} &= \mathbf{d}_{n+1} + \mathbf{q}_n - \mathbf{d}_n - \mathbf{d}_{n+3} - \mathbf{d}_{n+6}, \\ \mathbf{Y}_{3,n} &= \mathbf{d}_{n+2} + \mathbf{r}_n - \mathbf{d}_{n+4} - \mathbf{p}_n, & \mathbf{Y}_{4,n} &= \mathbf{d}_{n+3} + \mathbf{v}_n + \mathbf{j}_n - \mathbf{d}_{n+5} - \mathbf{q}_n.\end{aligned}\quad (72)$$

It follows that $\mathbf{X}_{i,n}$ satisfies the ultradiscretized expression of the original type D_6 ((55) with all $b_i = 1$), which can be obtained from the (max,+) equations (70), as shown below.

Lemma 4.1 *The quantities $\mathbf{X}_{j,n}$ in (71) satisfy the following system of (max,+) equations:*

$$\begin{aligned}\mathbf{X}_{1,n+1} + \mathbf{X}_{1,n} &= [\mathbf{X}_{2,n}]_+, \\ \mathbf{X}_{2,n+1} + \mathbf{X}_{2,n} &= [\mathbf{X}_{1,n+1} + \mathbf{X}_{3,n}]_+, \\ \mathbf{X}_{3,n+1} + \mathbf{X}_{3,n} &= [\mathbf{X}_{2,n+1} + \mathbf{X}_{4,n}]_+, \\ \mathbf{X}_{4,n+1} + \mathbf{X}_{4,n} &= [\mathbf{X}_{5,n} + \mathbf{X}_{6,n} + \mathbf{X}_{3,n+1}]_+, \\ \mathbf{X}_{5,n+1} + \mathbf{X}_{5,n} &= [\mathbf{X}_{4,n+1}]_+, \\ \mathbf{X}_{6,n+1} + \mathbf{X}_{6,n} &= [\mathbf{X}_{4,n+1}]_+, \end{aligned} \quad (73)$$

where $[a]_+ = \max(a, 0)$.

Given arbitrary initial values $(\mathbf{X}_{1,0}, \mathbf{X}_{2,0}, \mathbf{X}_{3,0}, \mathbf{X}_{4,0}, \mathbf{X}_{5,0}, \mathbf{X}_{6,0})$, the quantities $\mathbf{X}_{j,n}$ (and $\mathbf{Y}_{i,n}$) are periodic with period 6 for $1 \leq i \leq 4$ and $1 \leq j \leq 6$.

Proof The (max,+) equations in (73) arise from (70) by rearranging the d-vectors so that they can be expressed in terms of quantities $\mathbf{X}_{j,n}$ in (71). Alternatively, these equations can be derived from the exchange relations of type D_6 cluster map, φ_{D_6} , (25), by considering the d- vectors of cluster variables x_j . Since φ_{D_6} is periodic with period 6, it follows that each quantity $\mathbf{X}_{j,n}$ inherits this periodicity. $\square$

Note that for $1 \leq j \leq 4$, each $\mathbf{Y}_{j,n}$ satisfies the same periodicity since it can be expressed in terms of $\mathbf{X}_{i,n}$ for all $1 \leq i \leq 6$. The periodicity of $\mathbf{X}_{i,n}$ (or $\mathbf{Y}_{j,n}$) is essential in determining the degree growth of the d-vectors of tau functions as demonstrated in the proof of the following statement.

Theorem 4.2 *Let $\mathcal{T}$ be a linear operator which shifts $n \rightarrow n + 1$. The d-vectors $\mathbf{e}_n, \mathbf{d}_n, \mathbf{f}_n$ and $\mathbf{g}_n$, which solve the system of equations (70), satisfy the following linear difference equations*

$$\mathcal{L}\mathbf{r}_n = (\mathcal{T}^6 - 1)(\mathcal{T}^5 - 1)(\mathcal{T} - 1)\mathbf{r}_n = 0 \quad (74)$$

where $\mathbf{r}_n = \mathbf{e}_n, \mathbf{d}_n, \mathbf{f}_n, \mathbf{g}_n$. For the generated tau functions, the leading order of degree growth of their denominators is given by

$$\begin{aligned}\mathbf{d}_n &= \frac{n^2}{60}\mathbf{a} + O(n), \quad \mathbf{v}_n = \frac{n^2}{60}\mathbf{a} + O(n), \quad \mathbf{j}_n = \frac{n^2}{60}\mathbf{a} + O(n) \\ \mathbf{p}_n &= \frac{n^2}{30}\mathbf{a} + O(n), \quad \mathbf{q}_n = \frac{n^2}{30}\mathbf{a} + O(n), \quad \mathbf{r}_n = \frac{n^2}{30}\mathbf{a} + O(n), \end{aligned} \quad (75)$$

where $\mathbf{a} = (2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2)^T$.

Proof The quantity $\mathbf{X}_{1,n}$ can be written in terms of the shift operator as following

$$\mathbf{X}_{1,n} = (\mathcal{T}^6 - \mathcal{T}^5 - \mathcal{T} + 1)\mathbf{d}_n \quad (76)$$

Since $\mathbf{X}_{1,n}$ exhibits periodicity of period 6, shown in Lemma 4.1, it follows that the quantity (76) yields a characteristic equation in $\mathcal{T}$,

$$(\mathcal{T}^6 - 1)\mathbf{X}_{1,n} = (\mathcal{T} - 1)^3 \left(\sum_{i=0}^5 \mathcal{T}^i \right) \left(\sum_{i=0}^4 \mathcal{T}^i \right) \mathbf{d}_n = 0. \quad (77)$$

Solving this equation provides the general solution for the $\mathbf{d}_n$ whose leading order term is n^2 with some constant vector $\mathbf{a}$,

$$\mathbf{d}_n = \mathbf{a}n^2 + O(n). \quad (78)$$

We now consider the relation (77) once again to determine the coefficient constant of the leading term of $\mathbf{d}_n$. The expression can be rewritten in the form $(\mathcal{T}^6 - 1)(\mathcal{T}^5 - 1)(\mathcal{T} - 1)\mathbf{d}_n$ which yields the following relation

$$\begin{aligned} (\mathcal{T}^6 - 1)(\mathcal{T}^5 - 1)(\mathcal{T} - 1)\mathbf{d}_n &= 0 \\ \implies (\mathcal{T}^6 - 1)(\mathcal{T}^5 - 1)\mathbf{d}_{n+1} &= (\mathcal{T}^6 - 1)(\mathcal{T}^5 - 1)\mathbf{d}_n. \end{aligned} \quad (79)$$

Substituting the general solution into this relation gives

$$(\mathcal{T}^6 - 1)(\mathcal{T}^5 - 1)\mathbf{d}_n = 60\mathbf{a}. \quad (80)$$

Evaluating the sequence of $\mathbf{d}_n$, obtained from the (max,+) relations (70) with initial d-vectors (69), and then substituting it into (80), we obtain

$$60\mathbf{a} = (2, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2)^T.$$

Thus the leading order term of $\mathbf{d}_n$ is:

$$\mathbf{d}_n = \frac{n^2}{60}\mathbf{a} + O(n)$$

which agrees with one of the results in (75).

In the case of d-vector $\mathbf{p}_n$, by using periodicity of quantity $\mathbf{X}_{2,n}$, $(\mathcal{T}^6 - 1)\mathbf{X}_{2,n} = 0$ and applying $(\mathcal{T}^5 - 1)(\mathcal{T} - 1)$ to both sides, one finds

$$(\mathcal{T}^6 - 1)(\mathcal{T}^5 - 1)(\mathcal{T} - 1)\mathbf{p}_n = 0$$

where $\mathbf{d}_{n+2}$ and $\mathbf{d}_{n+5}$ vanish since the $\mathbf{d}_n$ satisfy the relation (79). By using a similar argument above, one can show that the d-vectors $\mathbf{q}_n$ and $\mathbf{r}_n$ satisfy the same linear relation of $\mathcal{T}$ in (79).

As for the rest of d-vectors $\mathbf{v}_n$ and $\mathbf{j}_n$, we first consider the last two (max,+) relations in (70). Subtracting these relations then yields the following expression

$$(\mathcal{T} - 1)\mathbf{v}_n = (\mathcal{T} - 1)\mathbf{j}_n. \quad (81)$$

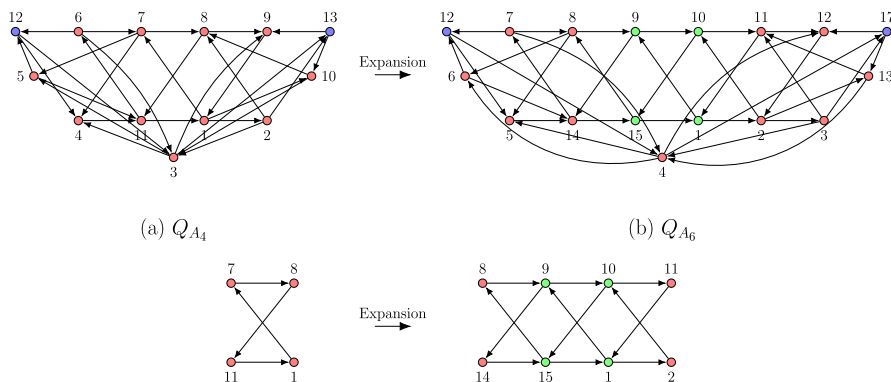


Fig. 9 Extension from Q_{A_4} to Q_{A_6} . Green nodes are new vertices which are inserted in Q_{A_4} in the form of local expansion

From the periodicity $(\mathcal{T}^6 - 1)\mathbf{Y}_{4,n} = 0$, the same procedure used in the case of $\mathbf{p}_n$ gives

$$(\mathcal{T}^6 - 1)(\mathcal{T}^5 - 1)(\mathbf{v}_{n+1} + \mathbf{j}_{n+1} - \mathbf{v}_n - \mathbf{j}_n) = 0. \quad (82)$$

Then substituting the identity (81), we find that both $\mathbf{v}_n$ and $\mathbf{j}_n$ are solutions of the recurrence in (79).

For the rest of proof (determining the coefficients of leading order terms of d-vectors), we can take the same approach as for the case $\mathbf{d}_n$ and the required results follow. $\square$

Since the degree growth of each variable is quadratic, associated algebraic entropy vanishes, which leads to the conjecture that the deformed type D_6 map is a Liouville integrable map.

5 Local Expansion

The main result of the previous work [7] is that the quiver $Q_{A_{2N}}$, arising from the deformed type A_{2N} map via Laurentification, can be obtained by inserting a specific subquiver (referred to as a *local expansion*) into the quiver Q_{A_2} , which itself arises from the Laurentification of the deformed type A_2 map, as shown in Fig. 9. Here we take a similar approach to the type D_4 .

Comparing the quivers, $Q_{D_4}^{(2)}$ (Fig. 6 (b)) and Q_{D_6} (Fig. 8), one can see that the $Q_{D_4}^{(2)}$ can be extended to Q_{D_6} via a transformation that splits the nodes 3 and 5 into two nodes each, then inserts a subquiver that has the same structure as illustrated in Fig. 10, and relabels the nodes accordingly.

This expansion suggests that the cluster $\tilde{\mathbf{x}}$, associated with $Q_{D_4}^{(2)}$, extends to the type D_6 case by shifting the subindices $1 \leq i \leq 5$ of the $\tilde{x}_i$ by 1 ($\tilde{x}_i \rightarrow \tilde{x}_{i+1}$), while the remaining indices are shifted by 3 ($\tilde{x}_i \rightarrow \tilde{x}_{i+3}$). The new cluster variables $\tilde{x}_1, \tilde{x}_7, \tilde{x}_8, \tilde{x}_{12}$, corresponding to additional nodes, are then inserted into the cluster $\tilde{\mathbf{x}}$, resulting in the cluster $\hat{\mathbf{x}}$ (66), which is associated with Q_{D_6} . From these examples, one can infer

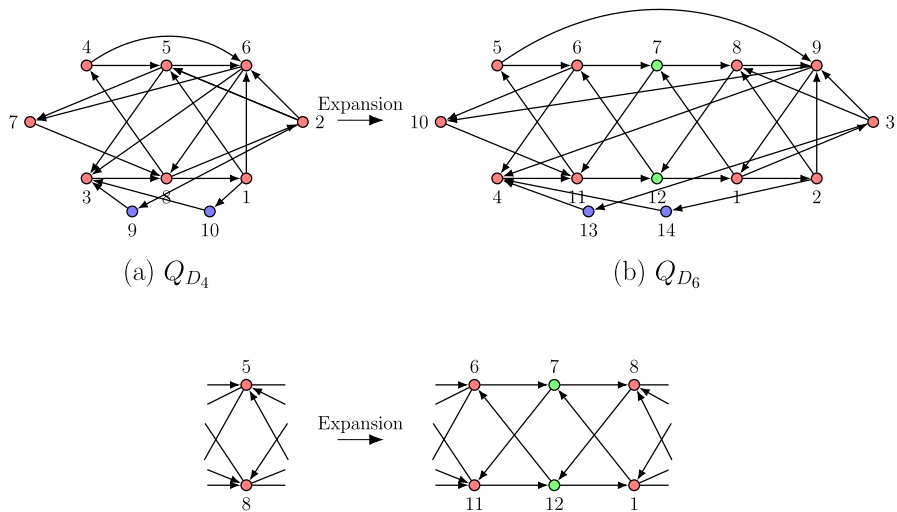


Fig. 10 Extension from Q_{D_4} to Q_{D_6}

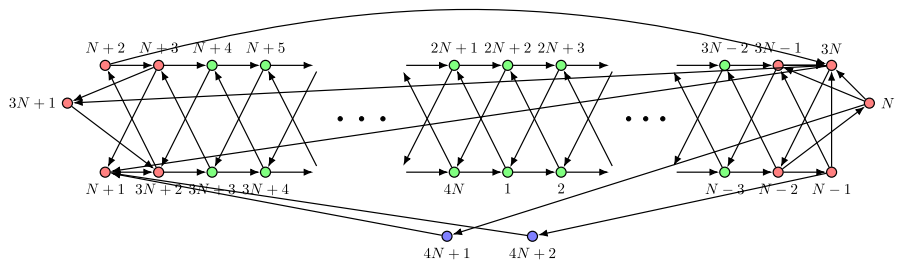


Fig. 11 (potential) quiver $Q_{D_{2N}}$ emerged via Laurentification of the deformed type D_{2N} map (83)

that applying this expansion recursively constructs a family of quivers with $4N$ nodes for $N \geq 3$ that arises from Laurentification of the deformed type D_{2N} map, $\tilde{\varphi}_{D_{2N}}$ whose iteration is given by

$$\begin{aligned}
 x_{1,n+1}x_{1,n} &= b_{2N-1}b_{2N} + x_{2,n} \\
 x_{2,n+1}x_{2,n} &= b_{2N-1}b_{2N} + x_{1,n+1}x_{3,n} \\
 &\vdots \\
 x_{N-2,n+1}x_{N-2,n} &= b_{2N-1}b_{2N} + x_{N-3,n+1}x_{N-1,n}x_{N,n} \\
 x_{N-1,n+1}x_{N-1,n} &= b_{2N-1} + x_{N-2,n+1} \\
 x_{N,n+1}x_{N,n} &= b_{2N} + x_{N-2,n+1}
 \end{aligned} \tag{83}$$

Let us denote the initial cluster variables, corresponding to the nodes of the quiver, by the following tau-functions,

$$\tilde{\mathbf{x}}_{2N} = (\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_{4N}) = (\xi_{N-2,0}, \dots, \xi_{1,0}, r_0, s_0, \tau_0, \tau_1, \dots, \tau_{2N}, \rho_0, \eta_{1,0}, \dots, \eta_{N-2,0}) \quad (84)$$

Recall that the variable transformations for the x variables in the type D_4 case are given by

$$x_1 = \frac{\tilde{x}_7 \tilde{x}_3}{\tilde{x}_6 \tilde{x}_4}, \quad x_2 = \frac{\tilde{x}_8}{\tilde{x}_6 \tilde{x}_5}, \quad x_3 = \frac{\tilde{x}_1 \sigma_0}{\tilde{x}_6}, \quad x_4 = \frac{\tilde{x}_2}{\tilde{x}_6 \sigma_0}$$

We then shift the indices of x_j for $j = 3, 4$ by 1 ($j \rightarrow j+1$), while at the same time shifting the indices of the $\tilde{x}_i$ variables, according to the cluster extension described above. After this shift, we define new variables

$$x_3 = \frac{\tilde{x}_{12}}{\tilde{x}_9 \tilde{x}_7}, \quad x_4 = \frac{\tilde{x}_1}{\tilde{x}_9 \tilde{x}_8}$$

By including this, we have the following variable transformation,

$$x_1 = \frac{\tilde{x}_{10} \tilde{x}_4}{\tilde{x}_9 \tilde{x}_5}, \quad x_2 = \frac{\tilde{x}_{11}}{\tilde{x}_9 \tilde{x}_6}, \quad x_3 = \frac{\tilde{x}_{12}}{\tilde{x}_9 \tilde{x}_7}, \quad x_4 = \frac{\tilde{x}_1}{\tilde{x}_9 \tilde{x}_8}, \quad x_5 = \frac{\tilde{x}_2 \sigma_0}{\tilde{x}_9}, \quad x_6 = \frac{\tilde{x}_3}{\tilde{x}_9 \sigma_0}$$

which matches (63). Then, as above, one can deduce the structure of the variable transformation (shown in (85)) that brings the deformed type D_{2N} map (83) into the form of cluster map defined by the cluster mutations in cluster algebra, equipped with the cluster $\tilde{\mathbf{x}}_{2N}$ and the quiver $Q_{D_{2N}}$.

$$\begin{aligned} x_1 &= \frac{\tilde{x}_{3N+1} \tilde{x}_{N+1}}{\tilde{x}_{3N} \tilde{x}_{N+2}}, \quad x_2 = \frac{\tilde{x}_{3N+2}}{\tilde{x}_{3N} \tilde{x}_{N+3}}, \quad \dots, \quad x_N = \frac{\tilde{x}_{4N}}{\tilde{x}_{3N} \tilde{x}_{2N+1}}, \\ x_{N+1} &= \frac{\tilde{x}_1}{\tilde{x}_{3N} \tilde{x}_{2N+2}}, \quad \dots, \quad x_{2N-2} = \frac{\tilde{x}_{N-2}}{\tilde{x}_{3N} \tilde{x}_{3N-1}}, \quad x_{2N-1} = \frac{\tilde{x}_{N-1} \sigma_0}{\tilde{x}_{3N}}, \\ x_{2N} &= \frac{\tilde{x}_N}{\tilde{x}_{3N} \sigma_0} \end{aligned} \quad (85)$$

As for the associated symplectic coordinates, we consider the symplectic structure in the cluster algebra of Dynkin type D_{2N} . Let $e_m = (0, \dots, 0, \underbrace{1}_{m^{\text{th}} \text{ position}}, 0, \dots, 0)$

be standard basis vectors for $1 \leq m \leq 2N$. The exchange matrix corresponding to Dynkin type D_{2N} , written in the form,

$$B_{D_{2N}} = \begin{pmatrix} 0 & 1 & & & & & & & \\ -1 & & \ddots & & & & & & \\ & \ddots & & 0 & 1 & & & & \\ & & \ddots & -1 & 0 & 1 & 0 & 0 & \\ & & & & -1 & 0 & 1 & 1 & \\ & & & & & 0 & -1 & 0 & 0 \\ & & & & & & 0 & -1 & 0 & 0 \end{pmatrix}$$

has rank $2N - 2 < 2N$, with kernel $\ker B_{D_{2N}}$ consisting of two linearly independent vectors, $\mathbf{u}_j = (u_1^{(j)}, u_2^{(j)}, \dots, u_{2N}^{(j)})$ for $j = 1, 2$, namely,

$$\mathbf{u}_1 = e_{2N} + \sum_{k=1}^{N-1} e_{2k-1}, \quad \mathbf{u}_2 = \sum_{k=1}^N e_{2k-1}. \quad (86)$$

and image $\operatorname{im} B_{D_{2N}}$ spanned by $2N - 2$ linearly independent vectors $\mathbf{v}_j = (v_1^{(j)}, v_2^{(j)}, \dots, v_{2N}^{(j)})$ of $\operatorname{im} B_{D_{2N}}$, for $1 \leq j \leq 2N - 2$, specifically,

$$\mathbf{v}_1 = e_2, \quad \mathbf{v}_l = e_{l+1} - e_{l-1} \quad (2 \leq l \leq 2N - 3), \quad \mathbf{v}_{2N-2} = e_{2N-1} + e_{2N} - e_{2N-3}. \quad (87)$$

Then null distribution of associated presymplectic form, ω ,

$$\omega = \sum_{i < j}^{2N} \frac{b_{ij}}{x_i x_j} dx_i \wedge dx_j$$

is generated by the two vector fields $\mathbf{z}_1 = x_{2N} \partial_{x_{2N}} + \sum_{k=1}^{N-1} x_{2k-1} \partial_{x_{2k-1}}$ and $\mathbf{z}_2 = \sum_{k=1}^N x_{2k-1} \partial_{x_{2k-1}}$. Therefore the corresponding symplectic coordinates are given by

$$\begin{aligned} y_1 &= x_2, \\ y_i &= x_{i+1}/x_{i-1} \quad \text{for } 2 \leq i \leq 2N - 3, \\ y_{2N-2} &= x_{2N-1}x_{2N}/x_{2N-3} \end{aligned}$$

and they are invariant under the flow generated by the vector fields $\mathbf{z}_1$ and $\mathbf{z}_2$. By substituting the x_i (85) into the reduced coordinates, we have

$$\begin{aligned} y_1 &= \frac{\tilde{x}_{3N+2}}{\tilde{x}_{3N}\tilde{x}_{N+3}}, \quad y_2 = \frac{\tilde{x}_{3N+3}\tilde{x}_{N+2}}{\tilde{x}_{N+4}\tilde{x}_{3N+1}\tilde{x}_{N+1}}, \quad y_3 = \frac{\tilde{x}_{3N+4}\tilde{x}_{N+3}}{\tilde{x}_{3N+2}\tilde{x}_{N+5}}, \dots, y_{N-1} = \frac{\tilde{x}_{4N}\tilde{x}_{2N-1}}{\tilde{x}_{4N-2}\tilde{x}_{2N+1}}, \\ y_N &= \frac{\tilde{x}_1\tilde{x}_{2N}}{\tilde{x}_{4N-1}\tilde{x}_{2N+2}}, \quad y_{N+1} = \frac{\tilde{x}_2\tilde{x}_{2N+1}}{\tilde{x}_{4N}\tilde{x}_{2N+3}}, \quad y_{N+2} = \frac{\tilde{x}_3\tilde{x}_{2N+2}}{\tilde{x}_1\tilde{x}_{2N+4}}, \dots, y_{2N-3} = \frac{\tilde{x}_{N-2}\tilde{x}_{3N-3}}{\tilde{x}_{N-4}\tilde{x}_{3N-1}}, \\ y_{2N-2} &= \frac{\tilde{x}_{N-1}\tilde{x}_N\tilde{x}_{3N-2}}{\tilde{x}_{3N}\tilde{x}_{N-3}} \end{aligned} \quad (88)$$

for $N \geq 4$. Through these variable transformations, one can verify that $Q_{D_{2N}}$ does indeed arise from the cluster algebra of type D_{2N} . Firstly, we consider the $(2N - 2) \times (2N - 2)$ coefficient matrix $\tilde{B} = (\tilde{b}_{ij})$ of the symplectic form $\hat{\omega}$,

$$\hat{\omega} = \sum_{i < j} \frac{\hat{b}_{ij}}{y_i y_j} dy_i \wedge dy_j$$

defined on the reduced space $\mathbf{y} = (y_1, \dots, y_{2N})$, that satisfies the pull-back relation $\pi^* \hat{\omega} = \omega$. Following the statement in Theorem 2.3, for $N \geq 4$, we define the $2N \times 2N$

matrix M , constructed from the vectors in $\text{im } B_{D_{2N}}$ (87) and in $\ker B_{D_{2N}}$ (86), as shown below

$$M = \begin{pmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \vdots \\ \mathbf{v}_{N-2} \\ \mathbf{u}_1 \\ \mathbf{u}_2 \end{pmatrix}. \quad (89)$$

The reduced matrix is then obtained via the relation (21),

$$M^{-T} B_{D_{2N}} M^{-1} = \begin{pmatrix} \hat{B} & 0 \\ 0 & 0 \end{pmatrix},$$

yielding the $2N - 2 \times 2N - 2$ skew-symmetric matrix whose entries are given by

$$\hat{b}_{i,j} = \begin{cases} 1 & i < j, i = 2m - 1, j = 2n, \quad 0 < n, m < N \\ 0 & j = 2n + 1, n \geq 1 \\ -1 & j < i, j = 2m - 1, i = 2n, \quad 0 < n, m < N \end{cases} \quad (90)$$

Secondly, we write the variable transformations (88) into the form $y_j = \tilde{\mathbf{x}}^{\mathbf{w}_j}$. Setting that the standard basis $\tilde{e}_i$ for $1 \leq i \leq 4N$, the vector $\mathbf{w}_j$ can be written as

$$\begin{aligned} \mathbf{w}_1 &= \tilde{e}_{3N+2} - \tilde{e}_{3N} - \tilde{e}_{N+3}, \\ \mathbf{w}_2 &= \tilde{e}_{3N+3} + \tilde{e}_{N+2} - \tilde{e}_{3N+1} - \tilde{e}_{N+4} - \tilde{e}_{N+1}, \\ \mathbf{w}_3 &= \tilde{e}_{3N+4} + \tilde{e}_{N+3} - \tilde{e}_{3N+2} - \tilde{e}_{N+5}, \\ &\vdots \\ \mathbf{w}_{N-1} &= \tilde{e}_{4N} + \tilde{e}_{2N-1} - \tilde{e}_{4N-2} - \tilde{e}_{2N+1} \\ \mathbf{w}_N &= \tilde{e}_1 + \tilde{e}_{2N} - \tilde{e}_{4N-1} - \tilde{e}_{2N+2} \\ \mathbf{w}_{N+1} &= \tilde{e}_2 + \tilde{e}_{2N+1} - \tilde{e}_{4N} - \tilde{e}_{2N+3} \\ \mathbf{w}_{N+2} &= \tilde{e}_3 + \tilde{e}_{2N+2} - \tilde{e}_1 - \tilde{e}_{2N+4} \\ &\vdots \\ \mathbf{w}_{2N-3} &= \tilde{e}_{N-2} + \tilde{e}_{3N-3} - \tilde{e}_{N-4} - \tilde{e}_{3N-1} \\ \mathbf{w}_{2N-2} &= \tilde{e}_{N-1} + \tilde{e}_N + \tilde{e}_{3N-2} - \tilde{e}_{3N} - \tilde{e}_{N-3} \end{aligned} \quad (91)$$

Let $\tilde{M}$ be $(2N - 2) \times 4N$ matrix given by

$$\tilde{M} = \begin{pmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \vdots \\ \mathbf{w}_{2N-2} \end{pmatrix} \quad (92)$$

The pull-back of the symplectic form

$$\hat{\pi}^* \hat{\omega} = \sum_{i < j}^{4N} \tilde{b}_{ij} d \log \tilde{x}_i \wedge d \log \tilde{x}_j \iff \tilde{B} = \tilde{M}^T \hat{B} \tilde{M} \quad (93)$$

gives rise to the $4N \times 4N$ skew-symmetric matrix $\tilde{B}$. Furthermore, appending the row corresponding to the frozen nodes at the bottom of the matrix $\tilde{B}$ yields the extended exchange matrix

$$\tilde{B}_{D_{2N}} = \begin{pmatrix} \begin{array}{cccccccc} 0 & 1 & & & & & & \\ -1 & & 0 & 1 & & & & \\ & 0 & 1 & & & & & \\ -1 & 0 & 1 & 1 & 0 & 0 & 0 & \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & \\ 0 & 0 & 0 & 0 & 0 & -1 & & \\ 0 & 0 & 0 & 0 & 0 & 1 & & \\ 0 & 0 & 0 & 1 & -1 & 0 & 1 & \end{array} & \begin{array}{cccccccc} 0 & 1 & 0 & -1 & & & & \\ & 1 & 0 & -1 & & & & \\ & 0 & 1 & 0 & -1 & 0 & 0 & \\ & 0 & 0 & 1 & 1 & 0 & 0 & \\ & 0 & 0 & 1 & 1 & 0 & 0 & \\ & 0 & 0 & 0 & -1 & 0 & 1 & \\ & 0 & 0 & 0 & 1 & 0 & -1 & \\ & 0 & 0 & 0 & 0 & 1 & 0 & -1 & \end{array} & \begin{array}{cc} -1 & \\ & 0 & -1 \\ & 1 & 0 & -1 \\ & & 1 & 0 & 1 \end{array} \\ \begin{array}{cccccccc} 0 & & & & & & & \\ -1 & & & & & & & \\ 0 & & & & & & & \\ 1 & & & & & & & \\ & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ & 1 & 0 & -1 & -1 & 0 & 0 & 0 \\ & & 1 & -1 & -1 & 1 & -1 & 0 \\ & & 0 & 0 & 0 & 0 & 0 & -1 \\ & & 0 & 0 & 0 & -1 & 1 & 0 \\ & & & & & & & 1 & 0 & -1 \end{array} & \begin{array}{cccccccc} -1 & 0 & 1 & 0 & 0 & & & \\ -1 & 0 & 1 & 0 & & & & \\ 0 & -1 & 0 & 1 & & & & \\ 0 & 0 & -1 & 0 & & & & \\ & & & & & & & 1 & 0 & -1 \end{array} \\ \begin{array}{cccccccc} 1 & & & & & & & \\ & 0 & 0 & -1 & 1 & 0 & & & 0 & 0 & 0 & 0 \\ & 0 & -1 & 0 & 1 & 0 & & & 0 & 0 & 0 & 0 \end{array} \end{pmatrix} \quad (94)$$

which represents the quiver $Q_{D_{2N}}$ (shown in Fig. 11).

6 Type D_{2N} Deformed Cluster Map

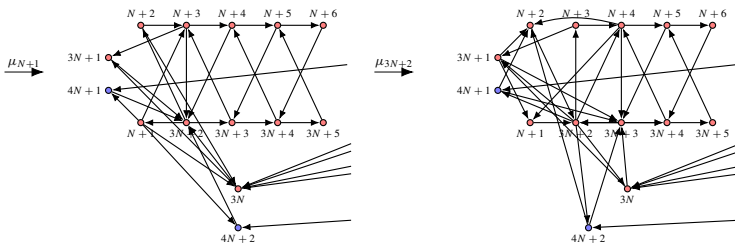
In the previous sections, we demonstrated that the deformed maps of type D_4 and D_6 can be described as the cluster maps on new cluster algebras of higher rank. Here we verify that the quiver $Q_{D_{2N}}$ (illustrated in Fig. 11) is mutation periodic under a specific sequence of mutations, which in turn gives rise to the cluster maps associated with deformed type D_{2N} maps via Laurentification.

Proposition 6.1 *For each $N \geq 2$, the quiver $Q_{D_{2N}}$ is mutation periodic,*

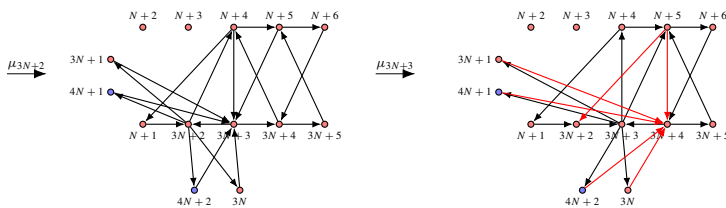
$$\mu_N \mu_{N-1} \cdots \mu_1 \mu_{4N} \mu_{4N-1} \cdots \mu_{3N+2} \mu_{N+1}(Q) = \rho(Q), \quad (95)$$

up to the permutation $\rho = (N-1, N)(N+1, N+2, \dots, 3N, 3N+1)$.

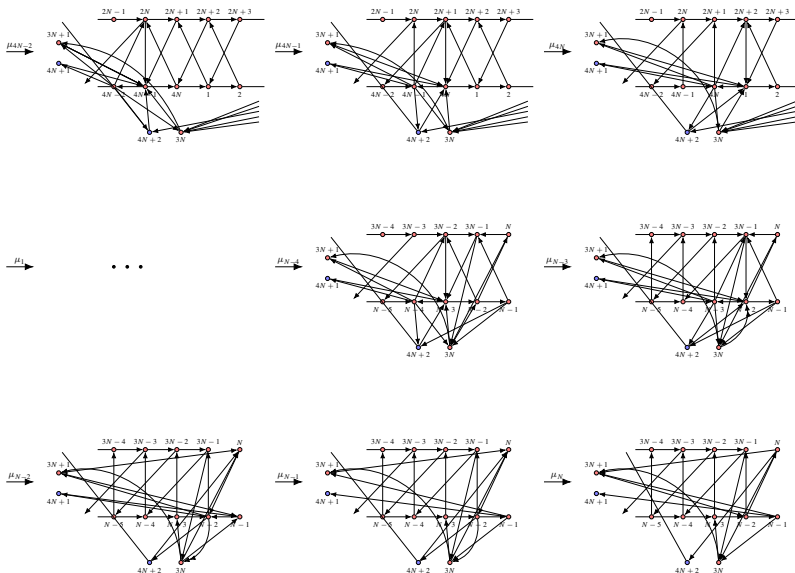
Proof Applying the mutation μ_{N+1} followed by the mutation μ_{3N+2} to the nodes of the quiver $Q_{D_{2N}}$ gives the following quiver.



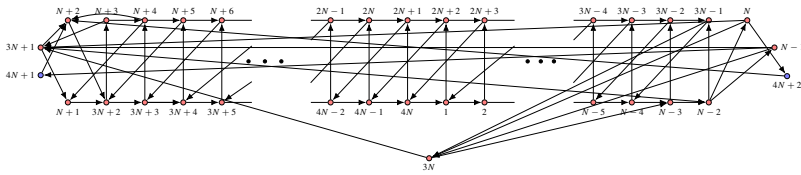
Let us consider the subquiver in the mutated quiver $\mu_{3N+2}\mu_1(Q_{D_{2N}})$, consisting of the next mutating node $3N+3$ along with all incident nodes that are connected to it. Upon applying the mutation μ_{3N+3} , we see that the node $3N+4$ is connected in a similar way as the previous mutating node. This resembles a rightward shift of the mutating node within the quiver as shown below.



Since the “ladder” shaped structural pattern of the quiver is extended toward the right end of the quiver, this transformation is repeated throughout the quiver. Thus, by applying the sequence of mutations $\mu_N\mu_{N-1}\cdots\mu_1\mu_{4N}\mu_{4N-1}\cdots\mu_{3N+4}$ sequentially, we see the following transformations within the quiver.



Thus after the sequence of mutations, the quiver yields following form.



By relabelling all of the nodes in the resulting quiver according to the permutation $\rho = (N-1, N)(N+1, N+2, \dots, 3N, 3N+1)$, the quiver returns to its initial form. Thus it is mutation periodic up to the permutation ρ i.e. $\mu_N \mu_{N-1} \cdots \mu_1 \mu_{4N} \mu_{4N-1} \cdots \mu_{3N+2} \mu_{N+1}(Q) = \rho(Q)$. $\square$

Therefore, since this composition of mutations is mutation periodic, one can define the cluster map $\psi = \rho^{-1} \mu_N \mu_{N-1} \cdots \mu_1 \mu_{4N} \mu_{4N-1} \cdots \mu_{3N+2} \mu_{N+1}$ such that its iteration on the initial cluster, $\tilde{\mathbf{x}}_{2N} = (\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_{4N})$, (84), generates new cluster variables via the following system of exchange relations,

$$\begin{aligned}
 \tilde{x}'_{N+1} \tilde{x}_{N+1} &= b_5 b_6 \tilde{x}_{3N} \tilde{x}_{N+3} + \tilde{x}_{3N+2} \\
 \tilde{x}'_{3N+2} \tilde{x}_{3N+2} &= b_5 b_6 \tilde{x}_{3N} \tilde{x}_{3N+1} \tilde{x}_{N+3} \tilde{x}_{N+4} + \tilde{x}'_{N+1} \tilde{x}_{3N+3} \tilde{x}_{N+2} \\
 \tilde{x}'_{3N+3} \tilde{x}_{3N+3} &= b_5 b_6 \tilde{x}_{3N} \tilde{x}_{3N+1} \tilde{x}_{N+4} \tilde{x}_{N+5} + \tilde{x}'_{3N+2} \tilde{x}_{3N+4} \\
 &\vdots \\
 \tilde{x}'_{4N-1} \tilde{x}_{4N-1} &= b_5 b_6 \tilde{x}_{3N} \tilde{x}_{3N+1} \tilde{x}_{2N} \tilde{x}_{2N+1} + \tilde{x}'_{4N-2} \tilde{x}_{4N} \\
 \tilde{x}'_{4N} \tilde{x}_{4N} &= b_5 b_6 \tilde{x}_{3N} \tilde{x}_{3N+1} \tilde{x}_{2N+1} \tilde{x}_{2N+2} + \tilde{x}'_{4N-1} \tilde{x}_1 \\
 \tilde{x}'_1 \tilde{x}_1 &= b_5 b_6 \tilde{x}_{3N} \tilde{x}_{3N+1} \tilde{x}_{2N+2} \tilde{x}_{2N+3} + \tilde{x}'_{4N} \tilde{x}_2 \\
 \tilde{x}'_2 \tilde{x}_2 &= b_5 b_6 \tilde{x}_{3N} \tilde{x}_{3N+1} \tilde{x}_{2N+3} \tilde{x}_{2N+4} + \tilde{x}'_1 \tilde{x}_3 \\
 &\vdots \\
 \tilde{x}'_{N-2} \tilde{x}_{N-2} &= b_5 b_6 \tilde{x}_{3N}^2 \tilde{x}_{3N+1} \tilde{x}_{3N-1} + \tilde{x}'_{N-3} \tilde{x}_{N-1} \tilde{x}_N \\
 \tilde{x}'_{N-1} \tilde{x}_{N-1} &= b_5 \tilde{x}_{3N} \tilde{x}_{3N+1} + \tilde{x}'_{N-2} \\
 \tilde{x}'_N \tilde{x}_N &= b_6 \tilde{x}_{3N} \tilde{x}_{3N+1} + \tilde{x}'_{N-2}.
 \end{aligned} \tag{96}$$

Upon setting the initial cluster as $\tilde{\mathbf{x}}_{2N} = (\xi_{N-2,0}, \dots, \xi_{1,0}, r_0, s_0, \tau_0, \tau_1, \dots, \tau_{2N}, \chi_0, \eta_{1,0}, \dots, \eta_{N-2,0})$ and denoting

$$\begin{aligned}
 \tilde{x}'_i &= \xi_{N-1-i,1} \quad (1 \leq i \leq N-2), \quad \tilde{x}'_{N-1} = s_1, \quad \tilde{x}'_N = r_1, \quad \tilde{x}'_{N+1} = \tau_{2N+1}, \\
 \tilde{x}'_{3N+2} &= \chi_1, \quad \tilde{x}'_j = \eta_{j-3N-1,1} \quad (3N+2 \leq j \leq 4N),
 \end{aligned} \tag{97}$$

the iteration of the cluster map is equivalent to the following system of recursion relations,

$$\begin{aligned}
 \tau_{n+2N+1}\tau_n &= b_5b_6\tau_{n+2N-1}\tau_{n+2} + \chi_n \\
 \chi_{n+1}\chi_n &= b_5b_6\tau_{n+2N-1}\tau_{n+2N}\tau_{n+2}\tau_{n+3} + \eta_{1,n}\tau_{n+2N+1}\tau_{n+1} \\
 \eta_{1,n+1}\eta_{1,n} &= b_5b_6\tau_{n+2N-1}\tau_{n+2N}\tau_{n+3}\tau_{n+4} + \chi_{n+1}\eta_{2,n} \\
 &\vdots \\
 \eta_{N-3,n+1}\eta_{N-3,n} &= b_5b_6\tau_{n+2N-1}\tau_{n+2N}\tau_{n+N-1}\tau_{n+N} + \eta_{N-4,n+1}\eta_{N-2,n} \\
 \eta_{N-2,n+1}\eta_{N-2,n} &= b_5b_6\tau_{n+2N-1}\tau_{n+2N}\tau_{n+N}\tau_{n+N+1} + \eta_{N-3,n+1}\xi_{N-2,n} \\
 \xi_{N-2,n+1}\xi_{N-2,n} &= b_5b_6\tau_{n+2N-1}\tau_{n+2N}\tau_{n+N+1}\tau_{n+N+2} + \eta_{N-2,n+1}\xi_{N-3,n} \\
 \xi_{N-3,n+1}\xi_{N-3,n} &= b_5b_6\tau_{n+2N-1}\tau_{n+2N}\tau_{n+N+2}\tau_{n+N+3} + \xi_{N-2,n+1}\xi_{N-4,n} \\
 &\vdots \\
 \xi_{1,n+1}\xi_{1,n} &= b_5b_6\tau_{n+2N-1}^2\tau_{n+2N}\tau_{n+2N-2} + \xi_{2,n+1}r_ns_n \\
 s_{n+1}r_n &= b_5\tau_{n+2N-1}\tau_{n+2N} + \xi_{1,n+1} \\
 r_{n+1}s_n &= b_6\tau_{n+2N-1}\tau_{n+2N} + \xi_{1,n+1}.
 \end{aligned} \tag{98}$$

Writing the variable transformations,

$$\begin{aligned}
 x_{1,n} &= \frac{\tau_{n+2N}\tau_n}{\tau_{n+2N-1}\tau_{n+1}}, \quad x_{2,n} = \frac{\chi_n}{\tau_{n+2N-1}\tau_{n+2}}, \quad x_{3,n} = \frac{\eta_{1,n}}{\tau_{n+2N-1}\tau_{n+3}}, \\
 \dots, x_{N,n} &= \frac{\eta_{N-2,n}}{\tau_{n+2N-1}\tau_{n+N}}, \quad x_{N+1,n} = \frac{\xi_{N-2,n}}{\tau_{n+2N-1}\tau_{n+N+1}}, \dots, \\
 x_{2N-2,n} &= \frac{\xi_{1,n}}{\tau_{n+2N-1}\tau_{n+2N-2}}, \quad x_{2N-1,n} = \frac{r_n}{\tau_{n+2N-1}}, \quad x_{2N,n} = \frac{s_n}{\tau_{n+2N-1}}
 \end{aligned} \tag{99}$$

the recursion relation reduce to the expressions

$$\begin{aligned}
 x_{1,n+1}x_{1,n} &= b_5b_6 + x_{2,n} \\
 x_{2,n+1}x_{2,n} &= b_5b_6 + x_{3,n}x_{1,n+1} \\
 &\vdots \\
 x_{2N-3,n+1}x_{2N-3,n} &= b_5b_6 + x_{2N-2,n}x_{2N-4,n+1} \\
 x_{2N-2,n+1}x_{2N-2,n} &= b_5b_6 + x_{2N-1,n}x_{2N,n}x_{2N-3,n+1} \\
 x_{2N-1,n+1}x_{2N-1,n} &= b_5 + x_{2N-2,n+1} \\
 x_{2N,n+1}x_{2N,n} &= b_6 + x_{2N-2,n+1}
 \end{aligned} \tag{100}$$

which correspond to the deformed type D_{2N} map. This shows that the deformed type D_{2N} can be expressed as a cluster map defined by cluster mutations in the cluster algebra whose initial seed consists of the cluster $\tilde{\mathbf{x}} = (\tilde{x}_1, \dots, \tilde{x}_{4N})$ and the quiver $\mathcal{Q}_{D_{2N}}$.

Since we have restored the Laurent property of the deformed map, we can follow the procedure described in the Sect. 4.2 to determine the degree growth of the map and thereby compute the corresponding algebraic entropy, which provides a criterion for detecting integrability.

7 Degree Growth of Deformed Type D_{2N} Map

In the previous section, we showed that one can construct a family of cluster maps from a generalized quiver that can be projected onto a reduced space where the resulting map is the deformed type D_{2N} map. Here we calculate the degree growth of the cluster map which correlates with the degree growth of the deformed map.

The cluster variables generated by iteration of the cluster map $\psi_{D_{2N}}$, starting from the initial cluster $\tilde{\mathbf{x}}_{2N} = (\xi_{N-2,0}, \dots, \xi_{1,0}, r_0, s_0, \tau_0, \tau_1, \dots, \tau_{2N}, \chi_0, \eta_{1,0}, \dots, \eta_{N-2,0})$, can be written in the form,

$$\begin{aligned} \xi_{N-1-i,n} &= \frac{P_n^{(i)}(\tilde{\mathbf{x}})}{\tilde{\mathbf{x}}^{\mathbf{h}_{N-1-i,n}}}, \quad r_n = \frac{P_n^{(N-1)}(\tilde{\mathbf{x}})}{\tilde{\mathbf{x}}^{\mathbf{r}_n}}, \quad s_n = \frac{P_n^{(N)}(\tilde{\mathbf{x}})}{\tilde{\mathbf{x}}^{\mathbf{s}_{i,n}}}, \\ \tau_n &= \frac{P_n^{(N+1)}(\tilde{\mathbf{x}})}{\tilde{\mathbf{x}}^{\mathbf{d}_n}}, \quad \chi_n = \frac{P_n^{(N+2)}(\tilde{\mathbf{x}})}{\tilde{\mathbf{x}}^{\mathbf{f}_n}}, \quad \eta_{i,n} = \frac{P_n^{(i+N+2)}(\tilde{\mathbf{x}})}{\tilde{\mathbf{x}}^{\mathbf{g}_{i,n}}} \end{aligned} \quad (101)$$

, for $1 \leq i \leq N-2$, where each degree vector appearing in the denominator is $4N \times 1$ vector (e.g. $\mathbf{d}_n = (h_{N-2,n}^{(N+1)}, h_{N-3,n}^{(N+1)}, \dots, h_{1,n}^{(N+1)}, r_n^{(N+1)}, s_n^{(N+1)}, d_n^{(N+1)}, \dots, d_{n+2N}^{(N+1)}, f_n^{(N+1)}, g_{1,n}^{(N+1)}, \dots, g_{N-2,n}^{(N+1)})$, which has initial data given by $4N \times 4N$ identity matrix,

$$(\mathbf{h}_{N-2,n}, \mathbf{h}_{N-3,n}, \dots, \mathbf{h}_{1,n}, \mathbf{r}_n, \mathbf{s}_n, \mathbf{d}_n, \dots, \mathbf{d}_{n+2N}, \mathbf{f}_n, \mathbf{g}_{1,n}, \dots, \mathbf{g}_{N-2,n}) = -I \quad (102)$$

Substituting these variables into the relations (98) and comparing the denominators on both sides gives following (max,+) relations,

$$\begin{aligned} \mathbf{d}_{n+2N+1} + \mathbf{d}_n &= \max(\mathbf{d}_{n+2N-1} + \mathbf{d}_{n+2}, \mathbf{f}_n), \\ \mathbf{f}_{n+1} + \mathbf{f}_n &= \max(\mathbf{d}_{n+2N-1} + \mathbf{d}_{n+2N} + \mathbf{d}_{n+2} + \mathbf{d}_{n+3}, \mathbf{g}_{1,n} + \mathbf{d}_{n+2N+1} + \mathbf{d}_{n+1}), \\ \mathbf{g}_{1,n+1} + \mathbf{g}_{1,n} &= \max(\mathbf{d}_{n+2N-1} + \mathbf{d}_{n+2N} + \mathbf{d}_{n+3} + \mathbf{d}_{n+4}, \mathbf{f}_{n+1} + \mathbf{g}_{2,n}) \\ &\vdots \\ \mathbf{g}_{N-3,n+1} + \mathbf{g}_{N-3,n} &= \max(\mathbf{d}_{n+2N-1} + \mathbf{d}_{n+2N} + \mathbf{d}_{n+N-1} + \mathbf{d}_{n+N}, \mathbf{g}_{N-4,n+1} + \mathbf{g}_{N-2,n}), \\ \mathbf{g}_{N-2,n+1} + \mathbf{g}_{N-2,n} &= \max(\mathbf{d}_{n+2N-1} + \mathbf{d}_{n+2N} + \mathbf{d}_{n+N} + \mathbf{d}_{n+N+1}, \mathbf{g}_{N-3,n+1} + \mathbf{h}_{N-2,n}), \\ \mathbf{h}_{N-2,n+1} + \mathbf{h}_{N-2,n} &= \max(\mathbf{d}_{n+2N-1} + \mathbf{d}_{n+2N} + \mathbf{d}_{n+N+1} + \mathbf{d}_{n+N+2}, \mathbf{g}_{N-2,n+1} + \mathbf{h}_{N-3,n}), \\ \mathbf{h}_{N-3,n+1} + \mathbf{h}_{N-3,n} &= \max(\mathbf{d}_{n+2N-1} + \mathbf{d}_{n+2N} + \mathbf{d}_{n+N+2} + \mathbf{d}_{n+N+4}, \mathbf{h}_{N-2,n+1} + \mathbf{h}_{N-4,n}), \\ &\vdots \\ \mathbf{h}_{1,n+1} + \mathbf{h}_{1,n} &= \max(2\mathbf{d}_{n+2N-1} + \mathbf{d}_{n+2N} + \mathbf{d}_{n+2N-2}, \mathbf{h}_{2,n+1} + \mathbf{r}_n + \mathbf{s}_n), \\ \mathbf{s}_{n+1} + \mathbf{r}_n &= \max(\mathbf{d}_{n+2N-1} + \mathbf{d}_{n+2N}, \mathbf{h}_{1,n+1}), \\ \mathbf{r}_{n+1} + \mathbf{s}_n &= \max(\mathbf{d}_{n+2N-1} + \mathbf{d}_{n+2N}, \mathbf{h}_{1,n+1}), \end{aligned} \quad (103)$$

Next we introduce quantities analogous to the tropical version of (63) as follows,

$$\begin{aligned}
 \mathbf{X}_{1,n} &= \mathbf{d}_{n+2N} + \mathbf{d}_n - \mathbf{d}_{n+2N-1} - \mathbf{d}_{n+1}, & \mathbf{X}_{2,n} &= \mathbf{f}_n - \mathbf{d}_{n+2N-1} - \mathbf{d}_{n+2}, \\
 \mathbf{X}_{3,n} &= \mathbf{g}_{1,n} - \mathbf{d}_{n+2N-1} - \mathbf{d}_{n+3}, & \mathbf{X}_{4,n} &= \mathbf{g}_{2,n} - \mathbf{d}_{n+2N-1} - \mathbf{d}_{n+4}, \\
 & \vdots & & \\
 \mathbf{X}_{N,n} &= \mathbf{g}_{N-2,n} - \mathbf{d}_{n+2N-1} - \mathbf{d}_{n+N}, & \mathbf{X}_{N+1,n} &= \mathbf{h}_{N-2,n} - \mathbf{d}_{n+2N-1} - \mathbf{d}_{n+N+1}, \\
 & \vdots & & \\
 \mathbf{X}_{2N-2,n} &= \mathbf{h}_{1,n} - \mathbf{d}_{n+2N-1} - \mathbf{d}_{n+2N-2}, & \mathbf{X}_{2N-1,n} &= \mathbf{r}_n - \mathbf{d}_{n+2N-1}, \\
 \mathbf{X}_{2N,n} &= \mathbf{s}_n - \mathbf{d}_{n+2N-1}.
 \end{aligned} \tag{104}$$

In this setting, the $(\max, +)$ relations above can be reformulated as follows,

$$\begin{aligned}
 \mathbf{X}_{1,n+1} + \mathbf{X}_{1,n} &= \max(\mathbf{X}_{2,n}, 0) \\
 \mathbf{X}_{2,n+1} + \mathbf{X}_{2,n} &= \max(\mathbf{X}_{3,n} + \mathbf{X}_{1,n+1}, 0) \\
 & \vdots \\
 \mathbf{X}_{2N-3,n+1} + \mathbf{X}_{2N-3,n} &= \max(\mathbf{X}_{2N-2,n} + \mathbf{X}_{2N-4,n+1}, 0) \\
 \mathbf{X}_{2N-2,n+1} + \mathbf{X}_{2N-2,n} &= \max(\mathbf{X}_{2N-1,n} + \mathbf{X}_{2N,n} + \mathbf{X}_{2N-3,n+1}, 0) \\
 \mathbf{X}_{2N-1,n+1} + \mathbf{X}_{2N-1,n} &= \max(\mathbf{X}_{2N-2,n+1}, 0) \\
 \mathbf{X}_{2N,n+1} + \mathbf{X}_{2N,n} &= \max(\mathbf{X}_{2N-2,n+1}, 0).
 \end{aligned} \tag{105}$$

These relations can be obtained by tropicalizing the exchange relations that define the original cluster map $\varphi_{D_{2N}}$. This fact leads to the following result.

Lemma 7.1 *Each quantity $\mathbf{X}_{i,n}$, which satisfies the tropical analogue of the deformed map $\tilde{\varphi}_{D_{2N}}$, (105), is periodic with period $2N$.*

Proof This can be shown by taking the same procedure which is used in Lemma 4.1. $\square$

Such a periodicity property of each $\mathbf{X}_{i,n}$ enables us to find an explicit formula for the growth of each degree vector, as shown below.

Theorem 7.1 *Given that the initial data of the degree vectors $\mathbf{d}_n, \mathbf{f}_n, \mathbf{g}_{i,n}, \mathbf{h}_{i,n}, \mathbf{s}_n, \mathbf{r}_n$ is specified by (102), the system of equations (103) has a solution, expressed as follows*

$$\begin{aligned}
 \mathbf{d}_n &= \frac{n^2}{8N^2-4N} \mathbf{a} + O(n), & \mathbf{r}_n &= \frac{n^2}{8N^2-4N} \mathbf{a} + O(n), & \mathbf{s}_n &= \frac{n^2}{8N^2-4N} \mathbf{a} + O(n) \\
 \mathbf{f}_{i,n} &= \frac{n^2}{4N^2-2N} \mathbf{a} + O(n), & \mathbf{h}_{i,n} &= \frac{n^2}{4N^2-2N} \mathbf{a} + O(n), & \mathbf{g}_{i,n} &= \frac{n^2}{4N^2-2N} \mathbf{a} + O(n)
 \end{aligned} \tag{106}$$

where $1 \leq i \leq N-2$ and $\mathbf{a} = (a_k)_{1 \leq k \leq 4N}$, where $a_l = 2$ for $l \in \{1, \dots, N-2\} \cup \{3N+2, \dots, 4N\}$ and $a_j = 1$ for $m = \{N-1, N\}$.

Proof By using the fact that $\mathbf{X}_{i,n} = \mathbf{X}_{i,n+2N}$ (periodicity of $\mathbf{X}_{i,n}$), we can obtain the linear relations for the degree vector $\mathbf{d}_n$ from the quantity $\mathbf{X}_{1,n}$ as follows,

$$\begin{aligned} (\mathcal{T}^{2N} - 1)(\mathbf{d}_{n+2N} + \mathbf{d}_n - \mathbf{d}_{n+2N-1} - \mathbf{d}_{n+1}) &= 0 \\ \iff (\mathcal{T}^{2N} - 1)(\mathcal{T}^{2N} - \mathcal{T}^{2N-1} - \mathcal{T} + 1) &= 0 \\ \iff (\mathcal{T} - 1)^3(\mathcal{T}^{2N-1} + \mathcal{T}^{2N-2} + \dots + \mathcal{T} + 1)(\mathcal{T}^{2N-2} + \mathcal{T}^{2N-3} + \dots + \mathcal{T} + 1) &= 0 \end{aligned}$$

The roots of this equation are $\exp(\frac{2k\pi i}{2N})$, $\exp(\frac{2k\pi i}{2N-1})$ and 1 with multiplicity 3. Thus the solution is of the form

$$\mathbf{d}_n = \mathbf{c}n^2 + O(n) \quad (107)$$

where $\mathbf{c}$ is a constant vector. The constant $\mathbf{c}$ can be determined by factorising the $\mathcal{T}$ equation in the following way,

$$\begin{aligned} (\mathcal{T}^{2N} - 1)(\mathcal{T}^{2N-1} - 1)(\mathcal{T} - 1)\mathbf{d}_n &= 0 \\ \iff (\mathcal{T}^{2N} - 1)(\mathcal{T}^{2N-1} - 1)\mathbf{d}_n &= (\mathcal{T}^{2N} - 1)(\mathcal{T}^{2N-1} - 1)\mathbf{d}_{n+1} \\ \implies (\mathcal{T}^{2N} - 1)(\mathcal{T}^{2N-1} - 1)\mathbf{d}_n &= (8N^2 - 4N)\mathbf{c} = \mathbf{a} \end{aligned}$$

With an inductive approach, one can deduce that the constant $\mathbf{a} = (a_k)_{1 \leq k \leq 4N}$, where $\mathbf{a} = (a_k)_{1 \leq k \leq 4N}$, where $a_l = 2$ for $l \in \{1, \dots, N-2\} \cup \{3N+2, \dots, 4N\}$ and $a_j = 1$ for $m = \{N-1, N\}$. The remaining degree vectors can be derived by using the periodicity property of the quantities $\mathbf{X}_{i,n}$ together with the relation for $\mathbf{d}_n$. For instance, applying the operator $\mathcal{L} = (\mathcal{T}^{2N} - 1)(\mathcal{T}^{2N-1} - 1)(\mathcal{T} - 1)$ to $\mathbf{X}_{2,n}$, we have

$$\begin{aligned} \mathcal{L}\mathbf{X}_{2,n} &= \mathcal{L}\mathbf{f}_n - \underbrace{\mathcal{L}\mathbf{d}_{n+2N-1}}_0 - \underbrace{\mathcal{L}\mathbf{d}_{n+2}}_0 = 0 \\ \implies \mathbf{f}_n &= \mathbf{c}_2n^2 + O(n) \end{aligned} \quad (108)$$

The constant vector $\mathbf{c}_2$ can be determined immediately from the $\mathbf{X}_{2,n}$.

$$\begin{aligned} (\mathcal{T}^{2N} - 1)(\mathcal{T}^{2N-1} - 1)\mathbf{f}_n &= (\mathcal{T}^{2N} - 1)(\mathcal{T}^{2N-1} - 1)(\mathbf{d}_{n+2} + \mathbf{d}_{n+2N-1}) \\ &= 2\mathbf{c} + O(n) \end{aligned} \quad (109)$$

Thus, this d-vector is expressed as $\mathbf{f}_n = \frac{1}{4N^2-2N}\mathbf{a}n^2 + O(n)$. Repeating this procedure to other quantities $\mathbf{X}_{i,n}$, we can acquire the desired results. $\square$

As a result, the degree growth of the cluster map $\psi_{D_{2N}}$ is quadratic, which implies that the associated algebraic entropy vanishes. Hence we arrive at the following conjecture.

Conjecture 7.1 *The discrete dynamics induced by the iteration of cluster maps $\psi_{D_{2N}}$ is a discrete integrable system.*

8 Conclusion and Future Work

In this paper, we considered one of the main results introduced in [12], namely, the deformation of a periodic cluster map associated with Dynkin type D_4 . We showed that the D_4 cluster map admits a two-parameter integrable deformation and then lifted via Laurentification to a cluster map $\psi_{D_4^{(2)}}$ in 8 dimensions, with 2 frozen variables, that is generated by the quiver $Q_{D_4}^{(2)}$, (shown in Fig. 6). We then demonstrated that the quiver corresponding to the deformed type D_6 map can be constructed from $Q_{D_4}^{(2)}$ by inserting "ladder-shaped" quiver the local expansion (Fig. 10), in a similar manner to that of [7]. As a result, we can construct a family of quivers (illustrated in Fig. 11) that generate two parametric cluster maps arising from deformed type D_{2N} via Laurentification.

In recent studies of the type D_4 case with Hone and Mase, we have found a distinguished cluster map $\hat{\psi}$, built from a composition of 6 mutations and a specific permutation applied to the same cluster algebra. This map preserves the same invariant function (first integral of $\psi_{D_4^{(2)}}$) which is also an elliptic curve as it has genus one. This means that each map acts as a translation by a point on the curve. Thus we expect these two to be independent of each other and hence describe the Mordell-Weil group of the corresponding elliptic surface of rank 2 like in the case of type A_3 (considered in [12]), which is under investigation.

In the paper [17], Masuda, Okubo and Tsuda demonstrated that the translation of the Weyl group, composed with R and other distinct reflections, gives rise to the discrete Painlevé equation q - P_{VI} . In recent studies, we have found correspondence between cluster map $\hat{\psi}$ and a certain birational map R that is one of the generating elements (reflection) of a Weyl group, $W(D_5^{(1)})$. We expect that there is a close relation between cluster algebra with quiver $Q_{D_4}^{(2)}$ and q - P_{VI} , which is subject to future work.

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Data Availability No new data were created or analysed in this study.

Declarations

Conflicts of Interest The author declare that there is no conflict of interest.

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